

Identification of CD133⁺ Intercellsomes in Intercellular Communication to Offset Intracellular Signal Deficit

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Abstract

CD133 (prominin 1) is widely viewed as a cancer stem cell marker in association with drug resistance and cancer recurrence. Herein we report that with impaired RTK-Shp2 Ras-Erk signaling, heterogeneous hepatocytes form clusters that manage to divide during liver regeneration. These hepatocytes are characterized by upregulated CD133 while negative for other progenitor cell markers. Pharmaceutical inhibition of proliferative signaling also induced CD133 expression in various cancer cell types, suggesting an inherent and common mechanism of stress response. Super-resolution and electron microscopy localize CD133 on intracellular vesicles that apparently migrate between cells, which we name “intercellsome”. Isolated CD133⁺ intercellsomes are enriched with mRNAs rather than miRNAs. Single-cell RNA sequencing reveals lower intracellular diversity (entropy) of mitogenic mRNAs in Shp2-deficient cells, which may be remedied by intercellular mRNA exchanges between CD133⁺ cells. CD133-deficient cells are more sensitive to proliferative signal inhibition in livers and intestinal organoids. These data suggest a mechanism of intercellular communication to compensate intracellular signal deficit in various cell types.

eLife assessment

This **important** study was designed to examine the bypass of Ras/Erk signaling defects that enable limited regeneration in a mouse model of hepatic regeneration. This hepatocyte proliferation is associated with the expression by groups of cells of mRNA-loaded CD133⁺ intracellular vesicles that mediate an intercellular signaling pathway that supports proliferation. These are new observations, supported by **convincing** data, that have broad significance to the fields of regeneration and cancer.

Liver regeneration is a widely used animal model to dissect mechanisms that drive cell proliferation *in vivo*, with the unique regenerative capacity conserved from animals to humans (Fausto *et al.*, 2006 [↗](#); Michalopoulos & DeFrances, 1997 [↗](#); Taub, 2004 [↗](#)). One frequently used experimental platform is partial hepatectomy (PHx) of two-thirds liver in rodents. Although re-entering cell cycle of quiescent hepatocytes in adult liver is believed to be a primary mechanism of liver regeneration, possible involvement of stem/progenitor cells has also been explored (Michalopoulos, 2017 [↗](#); Miyajima *et al.*, 2014 [↗](#)). However, conflicting data exist, regarding the source and contribution of putative stem/progenitor cells in liver regeneration and hepatocarcinogenesis following hepatic injuries (Kopp *et al.*, 2016 [↗](#); Michalopoulos, 2017 [↗](#); Miyajima *et al.*, 2014 [↗](#)). A CD133⁺ cell population was regarded as a source of progenitor cells that contribute to liver regeneration (Dorrell *et al.*, 2011 [↗](#); Miyajima *et al.*, 2014 [↗](#)), although the identity and mechanism are yet to be deciphered.

Detected immediately following PHx was rapid increase of growth factors in the liver and blood, including HGF, EGF, IL-6 and TNFα (Michalopoulos, 2017 [↗](#)), accompanied by upregulation of proliferative signaling events, especially the Ras-Erk pathway, in hepatocytes. Shp2/Ptpn11, an SH2-containing tyrosine phosphatase, is a key molecule that promotes signaling from receptor tyrosine kinases (RTKs) to the Ras-Erk pathway (Chan & Feng, 2007 [↗](#); Chen *et al.*, 2016 [↗](#); Grossmann *et al.*, 2010 [↗](#); Matozaki *et al.*, 2009 [↗](#); Tajan *et al.*, 2015 [↗](#)). Ablating Shp2 in hepatocytes attenuated hepatocyte proliferation following PHx and also suppressed RTK-driven hepatocarcinogenesis (Bard-Chapeau *et al.*, 2006 [↗](#); Chen *et al.*, 2021 [↗](#); Liu *et al.*, 2018 [↗](#)). Of note, liver regeneration was impaired but not blocked by removal of Shp2 or other pro-proliferative molecules in hepatocytes (Fausto *et al.*, 2006 [↗](#); Michalopoulos & DeFrances, 1997 [↗](#); Taub, 2004 [↗](#)). Thus, one critical question is why and how a small number of hepatocytes overcome the diminished mitogenic signaling, to re-gain proliferative capacity.

In this study, we found that in Shp2-deficient livers, actively dividing hepatocytes formed distinctive clusters and highly expressed CD133, but were negative for other stem/progenitor cell markers. The CD133⁺ cell colonies appeared transiently following PHx and disappeared quickly after completion of liver regeneration. This phenotype was also observed in MET-deficient livers following liver injuries. CD133 expression was upregulated by proliferative signal inhibition in a variety of cancer cell lines from different organs, indicating a commonly shared mechanism. Remarkably, we identified and isolated CD133⁺ intracellular vesicles, which were enriched with transcripts of immediate early responsive genes (IEGs) and migrated between tightly contacting cells in colonies. Single cell RNA sequencing (scRNA-seq) demonstrated that cells defective for proliferative signaling exhibited lower intracellular transcriptional diversity (entropy), which was remedied in CD133⁺ cells, potentially by converting intercellular heterogeneity into intracellular diversity through sharing of mRNAs. These data suggest a new function of CD133 in intercellular communication, by which normal and cancer cells endeavor to proliferate under proliferative signal deficit.

Results

Shp2 deficiency induces clustering of proliferating cells transiently in regenerating liver

We chose mouse liver regeneration to dissect mechanisms of cell proliferation in mammals under physiological context. In previous experiments, we showed that removal of Shp2 suppressed Ras-Erk signaling and hepatocyte division following PHx in mice (Bard Chapeau *et al.*, 2006). However, it was unclear why certain Shp2-negative hepatocytes managed to divide when proliferative

signaling was impaired in general. To address this question, we examined carefully the regenerating process in hepatocyte-specific Shp2 knockout (SKO, *Shp2^{flox/flox}; Alb-Cre*) livers. At 2 days after PHx, WT and SKO livers exhibited similar overall and histological phenotypes (**Figure 1–figure supplements 1A** and **1B**). Immunostaining for Ki67 showed significantly lower numbers of proliferating hepatocytes in SKO than WT livers, with no difference for non-parenchymal cells (NPCs) where Shp2 was not deleted (**Figures 1A**, **Figure 1–figure supplements 1C**, and **1D**). Intriguingly, we identified certain areas that were enriched with proliferating hepatocytes in SKO liver (**Figure 1A**; arrow). This peculiar proliferation pattern was especially cryptic, because these areas did not display severe inflammation or different phenotypes of hepatocytes except active proliferation (**Figure 1A**; right panel). Therefore, we searched extensively for biomarker molecules, which could distinguish the actively proliferating cells in the particular areas, including GFAP, EpCAM, CK19, CD133, Sox9, CD44 and α fetoprotein (AFP) (**Figure 1–figure supplements 1E–1H**). Of note, CD133 was the only molecule that was highly expressed in the areas enriched with proliferating hepatocytes (**Figure 1B**). The percentages of Ki67⁺ hepatocytes were significantly higher in CD133⁺ than CD133[−] areas in regenerating livers (**Figure 1C** and **1D**). These patchy areas appeared 2 days after PHx, and underwent expansion until 3 weeks, whereas WT livers did not show comparable CD133 expression in hepatocytes (**Figure 1E**). Liver/body weight ratios were similar between SKO and WT mice by 3 weeks (**Figure 1–figure supplement 1I**), suggesting that the active division of CD133⁺ cells could compensate the impaired proliferative capacity of CD133[−] hepatocytes in SKO liver.

The CD133⁺ hepatocytes also expressed high levels of E-cadherin and displayed tight cell-cell contact in the colonies (**Figures 2A**, **Figure 1–figure supplements 1J** and **1K**). CD133 was localized to the apical side of hepatocytes (**Figure 2A**), and was rarely detected in other types of cells located on their basal sides. At 3 weeks after PHx, the colonies appeared more distinctive even in H&E staining, which contained smaller hepatocytes with compact structure, albeit no difference observed for NPC types and the vasculature (**Figure 2B**). Immunostaining for sinusoidal endothelial cells and HGF did not show any unique features in the colonies otherwise (**Figure 1–figure supplements 1L** and **1M**). The distinct colonies merged into surrounding tissues 5 weeks after PHx, as evidenced by loss of CD133 expression and disappearance of clear colony edges (**Figure 2C**). Therefore, the CD133⁺ colonies of proliferating hepatocytes were composed transiently during liver regeneration.

Due to the compacting colony architecture 3 weeks after PHx, we observed clear difference in light scattering/refraction in unstained thick liver sections (**Figure 2D**), which allowed manual dissection of the colonies out of liver sections for analysis. Immunoblotting confirmed enrichment of CD133⁺ and E-cadherin⁺ cells but barely detected Shp2 protein in the isolated colonies, indicating their origin from Shp2-deleted hepatocytes (**Figures 2D** and **2E**). CD133 is known to be expressed in oval cells, which can differentiate into both hepatocytes and bile duct epithelial cells, similar to the potential of hepatoblasts (**Figure 2F**). However, we did not observe expression of other oval cell markers, including EpCAM, AFP and Sox9, in the colony lysates, (**Figure 2E**), consistent with the immunostaining data (**Figure 1–figure supplements 1E–1H**). Indeed, hepatocytes in the colonies did not exhibit oval cell like morphology throughout the regenerative process (**Figures 1A**, **1E**, **2A–2C**). CD133 expression in colonized hepatocytes was even higher than the embryonic liver at E17.5, which highly expressed AFP, a hepatoblast marker (**Figure 2E**). The molecular size of CD133 expressed in the colonies was slightly different from that in bile duct (**Figure 2E**), likely reflecting variations in glycosylation (Barkeer *et al*, 2018; Kemper *et al*, 2010). Further, the colony-forming cells expressed a mature hepatocyte marker HNF4 α , at similar levels to non colony areas (**Figure 2E**). These results indicate that the clustered CD133⁺ cells in SKO liver are not hepatoblasts or oval cells but are mature hepatocytes that acquire unique proliferation potential following PHx.

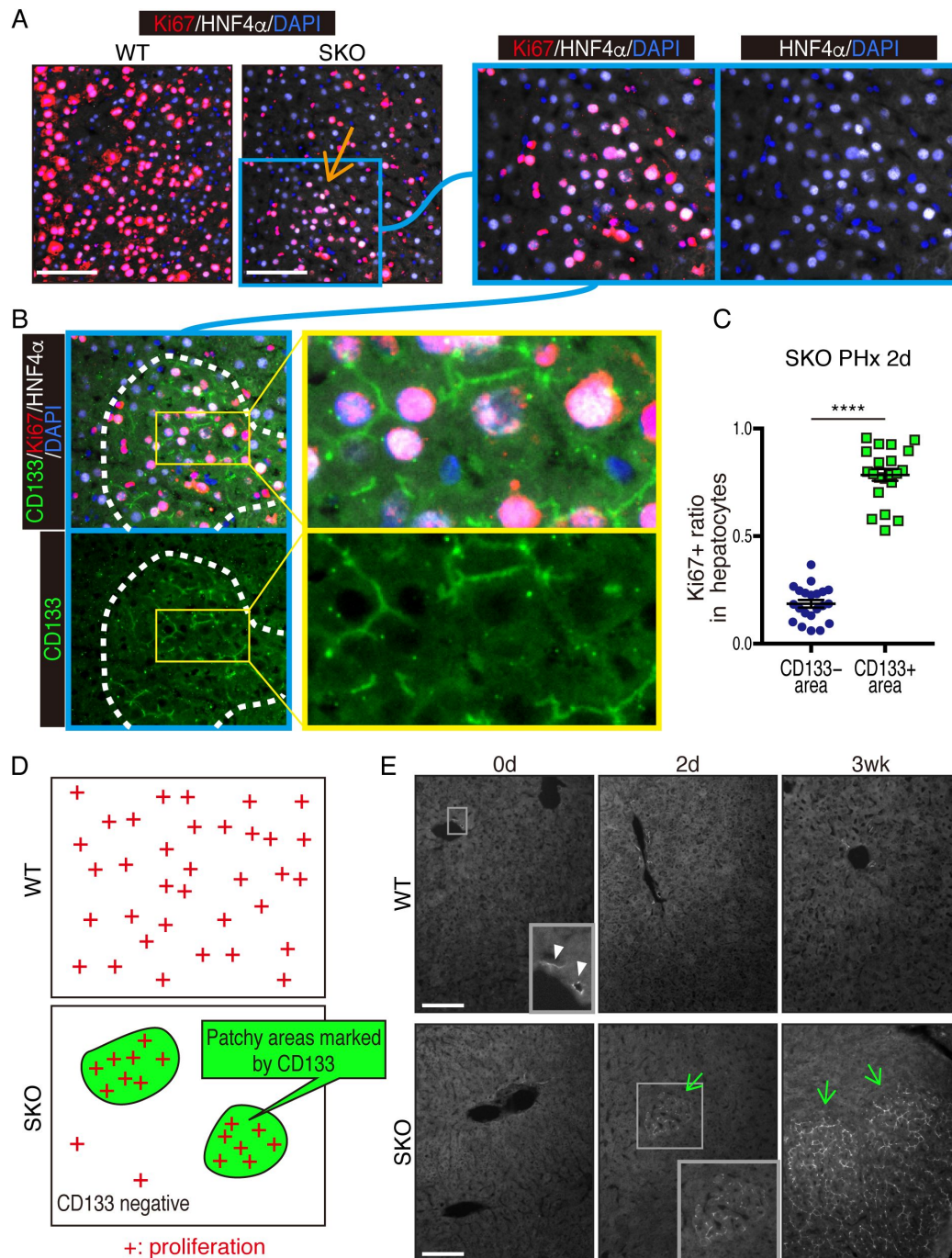


Figure 1.

Identification of a patchy hepatocyte proliferation pattern in Shp2-deficient (SKO) liver after PHx

(A and B) Immunofluorescence on liver tissue sections 2 days after PHx. HNF4α is a hepatocyte marker and Ki67 is a proliferation marker. Arrow in (A) points to an area enriched with proliferating hepatocytes. Dashed line in (B) shows an area with continuous CD133 expression.

(C) Quantification of proliferating rate in hepatocytes in CD133-positive and -negative areas in SKO livers 2 days after PHx. Each dot indicates one area. Data were collected from 3 mice. Means ± SEM are shown. **** $P < 0.0001$ (two-tailed unpaired t test).

(D) While WT hepatocytes proliferated at high frequency everywhere, proliferating hepatocytes in SKO liver were mostly located in patchy areas marked by CD133 expression.

(E) Immunofluorescence of CD133 on liver tissues at day 0 or 2, and 3 weeks (0d, 2d, 3 wk) after PHx. CD133⁺ hepatocyte clusters were only found in SKO livers after PHx (light green arrows). In WT livers, CD133 expression was only seen in bile duct epithelial cells (arrowheads).

Scale bars, 100 μm (A and E).

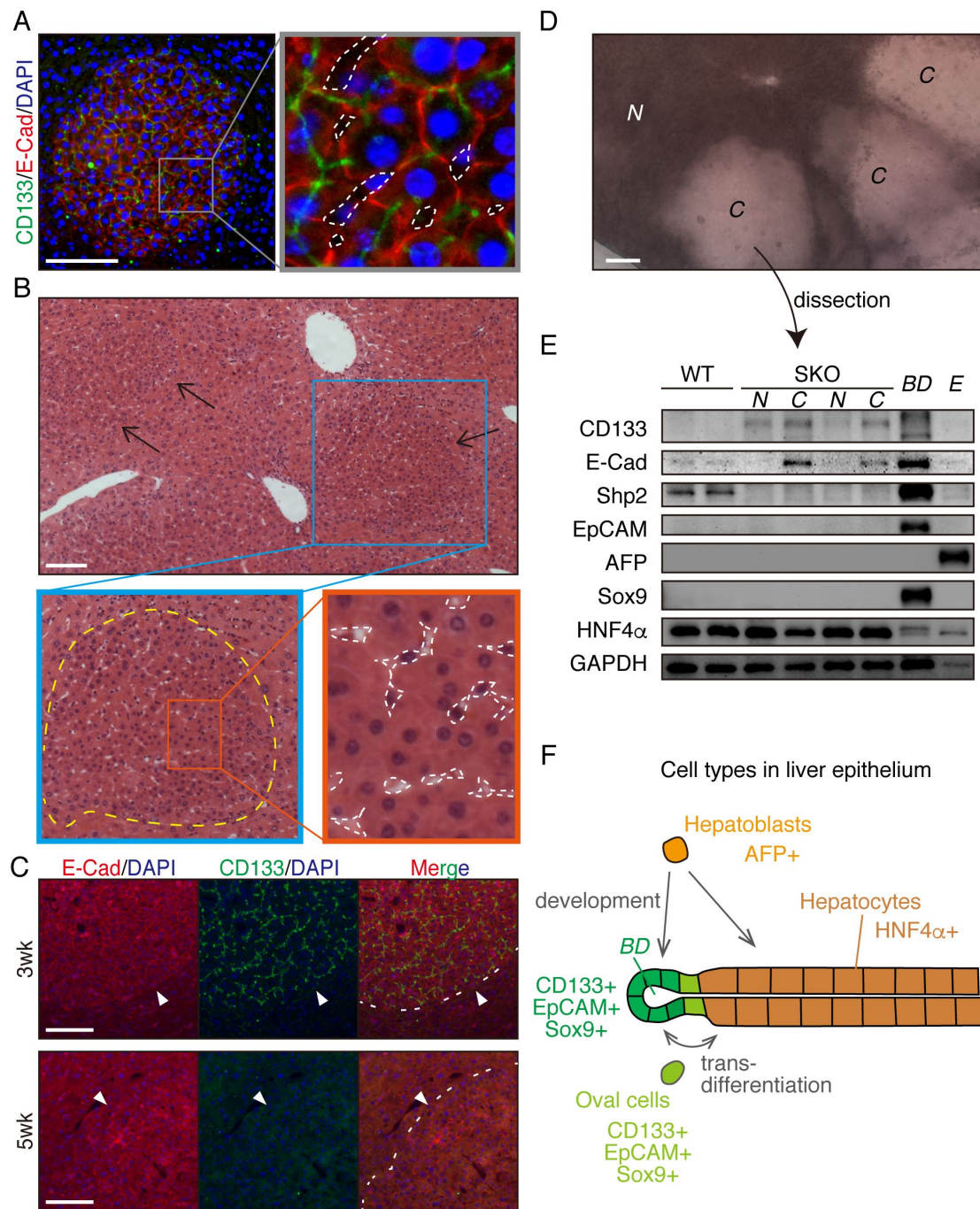


Figure 2.

CD133⁺ colonies represent a unique regeneration process in Shp2-deficient liver

(A) Representative image of CD133⁺ colonies shown by immunofluorescence on SKO liver tissues 3 weeks after PHx. Dashed lines, vasculatures.

(B) H&E staining of SKO liver sections 3 weeks after PHx. Yellow dashed line: boundary between the colony and surrounding tissue; white dashed lines: vasculatures.

(C) Immunofluorescence on SKO liver sections at indicated time points after PHx. Arrowheads and dashed lines indicate the boundaries between the colonies and surrounding tissues, which was clear at 3 weeks but disappeared at 5 weeks.

(D) Unstained thick tissue section of SKO liver 3 weeks after PHx.

(E) Colonies (C) and non-colony control areas (N) were dissected out from thick tissue sections of SKO livers 3 weeks after PHx, and analyzed by immunoblotting. Random areas from WT livers 3 weeks after PHx, bile duct (BD) and E17.5 liver (E) were used for comparison.

(F) Illustration of epithelial lineages in the liver. Scale bars, 100 μ m (A-D).

The proliferative cell colonies originate from heterogeneous mature hepatocytes

To determine the origin of these colonized hepatocytes by clonal tracing, we transfected vectors carrying a GFP and a *sleeping beauty* (SB) transposon (Ivics *et al.*, 1997 [\[1\]](#)) into liver by hydrodynamic tail vein injection (HTVi) (Liu *et al.*, 1999 [\[2\]](#); Zhang *et al.*, 1999 [\[3\]](#)), which labeled a few hepatocytes with GFP in a mosaic pattern (Figures 3A [\[4\]](#) and 3B [\[5\]](#)) (Mikkelsen *et al.*, 2003 [\[6\]](#); Yant *et al.*, 2000 [\[7\]](#)). We then performed PHx and analyzed regenerating livers 3 weeks later (Figures 3A [\[4\]](#) and 3B [\[5\]](#)). The resulting colonies in SKO livers contained both GFP⁺ and GFP⁻ cells (Figure 3B [\[5\]](#)), indicating multiple origins of hepatocytes. When a large GFP⁺ cell clone was located at a colony edge, we did not see recruitment of GFP⁻ cells, even though the surrounding hepatocytes were mostly GFP-negative (Figure 3C [\[6\]](#)). Thus, the colonies were expanded by proliferation of cells from inside rather than by continuous recruitment of surrounding hepatocytes (Figure 3-figure supplement 1A [\[7\]](#)). By measuring clone sizes of GFP⁺ hepatocytes inside and outside, we found that hepatocytes within the colonies divided much more actively than those located outside (Figures 3D [\[8\]](#) and 3E [\[9\]](#)), suggesting that clustering enabled hepatocytes to divide actively (Figure 3F [\[10\]](#)). The starting colony size at the fate determination stage (Figure 3F [\[10\]](#)) was estimated to be approximately 10 hepatocytes, based on the GFP⁺ clone/colony size ratios (Figures 3G [\[11\]](#) and Figure 3-figure supplement 1B [\[12\]](#)). The initial colony size should be large enough to include a variety of hepatocytes, signifying that the colony-forming potential does not require a special property of hepatocytes. The existence of mononuclear and binuclear cells (Duncan *et al.*, 2010 [\[13\]](#)) in the colonies 2 days after PHx further supports their heterogeneity in origin (Figure 3-figure supplement 1C [\[14\]](#)).

We reintroduced *WT* Shp2 together with GFP into a few Shp2-deficient hepatocytes by HTVi, and analyzed the mosaic liver tissues 2 days after PHx (Figure 3-figure supplement 2A [\[15\]](#)). Re-introducing Shp2 indeed restored the *WT* hepatocyte proliferation pattern in a cell autonomous manner (Figure 3-figure supplements 2B [\[16\]](#) and 2C [\[17\]](#)). Although the colonies were mainly composed of Shp2-negative hepatocytes (Figure 3-figure supplement 2D [\[18\]](#)), we observed Shp2⁺ cells inside the colonies, which were also CD133⁺ (Figure 3-figure supplement 2E [\[19\]](#)). Thus, Shp2 deficiency was not a requirement to join the colony, although Shp2⁺ cells disturbed colony morphology (Figure 3-figure supplement 2E [\[19\]](#)). Shp2-negative hepatocytes likely sent out paracrine signals to trigger clustering with surrounding hepatocytes, including Shp2⁺ cells in the neighborhood.

A similar pattern of colony formation was observed in cultures of primary hepatocytes isolated from SKO liver. Shp2-negative hepatocytes formed unique colonies of Ki67⁺ cells in vitro, resembling the observed clusters in liver sections (Figures 3H [\[20\]](#) and Figure 3-figure supplement 1D [\[21\]](#)), while proliferating hepatocytes were distributed rather randomly in the *WT* culture (Figure 3-figure supplement 1D [\[21\]](#)). E-cadherin expression in cell-cell contacts distinguished these colonies (Figure 3H [\[20\]](#)), featuring their similar properties to those in vivo. This phenotype was independent of colony sizes and therefore was not due to high cell density in the area (Figure 3H [\[20\]](#)). The colony formation in primary hepatocyte culture also confirms their origin from mature hepatocytes.

Cluster formation of dividing cells is a common mechanism under stresses

CCl4 is metabolized by hepatocytes around the central veins into a hyper-oxidative chemical that causes necrosis and apoptosis in the area (Forbes & Newsome, 2016 [\[22\]](#); Weber *et al.*, 2003 [\[23\]](#)). We examined CD133 induction in the liver following CCl4-induced injury, which involves cell death and inflammation, largely different microenvironment from the PHx model. At 2 days after CCl4 injection, we detected distinctive CD133⁺ hepatocyte clusters in SKO livers with higher proliferation rate than the surrounding areas (Figure 4A [\[24\]](#)). Although CCl4 also induced ductular

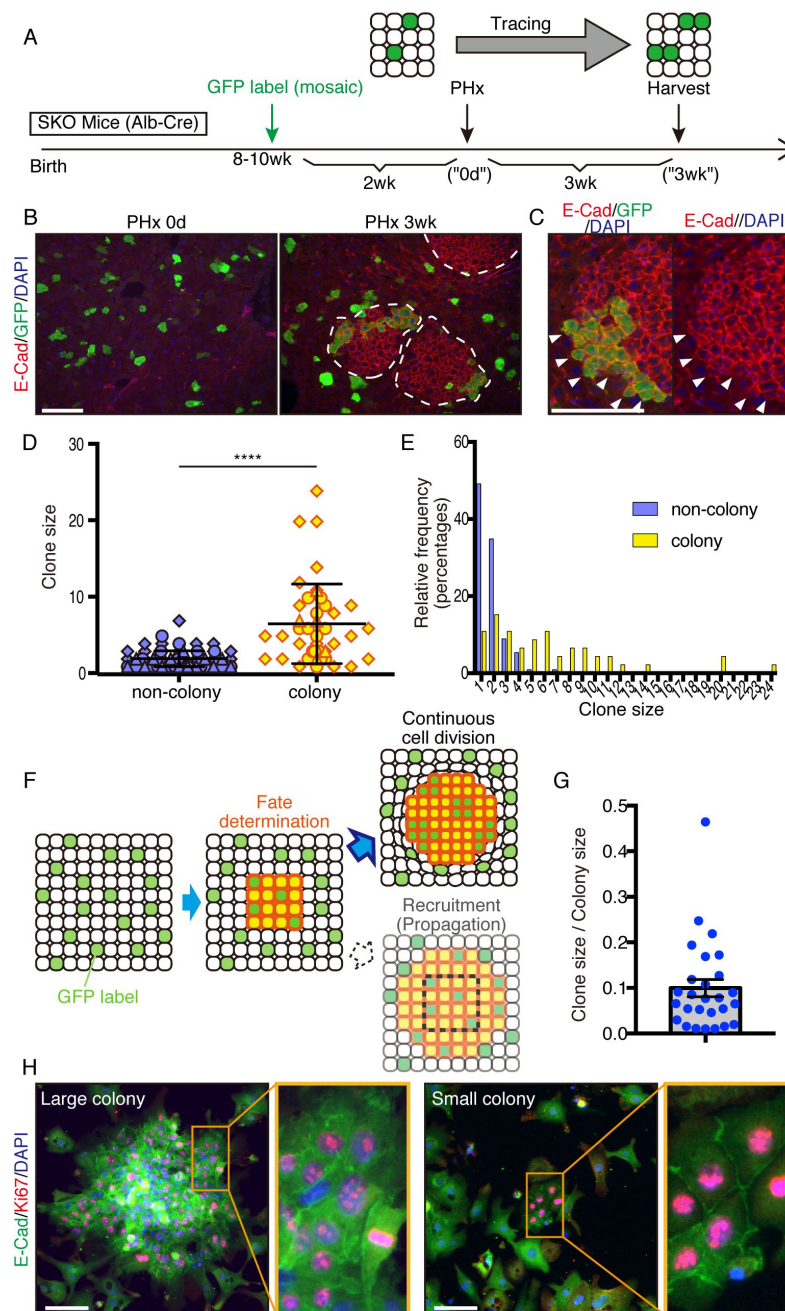


Figure 3.

Mitogenic signal deficiency induced tight clustering of cells that leads to continuous growth

(A) Schematic illustration of clonal tracing performed in SKO livers during regeneration.

(B) Liver tissues were examined before (PHx 0d) or 3 weeks after the surgery (PHx 3wk). E Cad shows the colony structures (arrows).

(C) GFP-labeled clone in a colony 3 weeks after PHx. Note that no unlabeled cells were detected on this edge (arrowheads).

(D) Cell numbers of GFP-labeled hepatocyte clones in colony and non-colony areas, counted on sections. Data were collected 3 weeks after PHx. Means $\pm$ SD are shown, $n = 3$, **** $P < 0.0001$ (Mann-Whitney test).

(E) Distribution analysis of clone sizes from (D).

(F) Clonal dynamics of colony-forming hepatocytes.

(G) Clone/colony size ratios to estimate the original colony size (see also Figure S2B). Mean $\pm$ SEM is shown.

(H) Immunofluorescence on the colonies (E-Cad⁺ clusters) in Shp2-deficient hepatocyte culture in vitro. Note the other non-colony cells with low proliferation rate, indicating the patchy proliferation in the colonies. See also Figure S2D.

Scale bars, 100 μ m (B, C and H).

reaction by CD133⁺ bile duct epithelial cells (Miyajima *et al.*, 2014). These areas were not associated with the CD133⁺ hepatocyte clusters (Figures 4A and 4B; arrowheads). Consistent with the PHx model, while bile duct epithelial cells were EpCAM⁺, CD133⁺ hepatocytes did not express EpCAM (Figure 4A). The CD133⁺ hepatocyte colonies appeared around the injured area, consistent with a notion of midlobular hepatocytes located next to a pericentral injury site (Wei *et al.*, 2021).

We further examined the effect of hepatocyte-specific deletion of HGF receptor MET in mice. Similar CD133⁺ proliferating hepatocyte clusters were induced by PHx or CCl₄ injection in MET-deficient livers (Figure 4C). These results suggest that CD133 induction is not restricted to a specific gene deletion or an injury model, but rather is a response to impaired proliferative signaling. Indeed, treatment of PLC liver cancer cells *in vitro* with a Shp2 inhibitor (SHP099) or Mek inhibitor (Trametinib), or a Shp2-targeting CRISPR vector, also upregulated CD133 expression (Figures 4D and 4E), indicating that CD133 induction is an inherent cellular response to impaired signaling of the RTK-Ras-Erk pathway. Upregulated CD133 expression was independent of other stem cell markers and thus did not reflect forced selection of a stem cell population (Figure 4D). Moreover, CD133 induction was rapid and dramatic in a manner that cannot be explained by selective expansion of pre existing CD133⁺ cells.

To determine if it is a widely conserved mechanism, we examined CD133 induction in a variety of cell lines derived from various organs and animals. Treatment with Shp2 or Mek inhibitor upregulated CD133 expression in mammary epithelial cells MCF10A (Figures 4F and 4G), glioblastoma cells GSC3028 and GSC23, insulinoma cells MIN6, colon adenocarcinoma cells MC38, and cervical cancer cells Hela as well as kidney fibroblasts COS7 (Figure 4H). Together, these results suggest that CD133 upregulation under proliferative signal deficit is a common mechanism in various cell types.

Upregulated CD133 is located on filament fibers connecting neighboring cells

Since the CD133⁺ cell clustering is apparently triggered by local intercellular signal, we examined expression of molecules known to mediate intercellular communication at short distance (Lander, 2007; Mikels & Nusse, 2006; Nusse, 2003; Rogers & Schier, 2011; Teleman *et al.*, 2001; Tsiarris & Aulehla, 2016). We detected significantly higher expression of Wnt7a, Wnt10a and Shh in SKO liver (Figure 3-figure supplement 2F). Consistently, porcupine (PORCN), a protein that resides in the ER and modifies Wnt for secretion (Clevers *et al.*, 2014; Mikels & Nusse, 2006; Nusse, 2003; Tammela *et al.*, 2017), was highly expressed in clustered hepatocytes (Figure 3-figure supplement 2G), suggesting a role of local Wnt signaling in colony assembly. PORCN expression was also strongly associated with the colonies in cultured primary SKO hepatocytes *in vitro* (Figure 3-figure supplement 2H), and PORCN protein amounts were higher in SKO than WT liver lysates 2 days after PHx (Figure 3-figure supplement 2I). Furthermore, the mRNA level of Wnt10a was drastically elevated in response to PHx in SKO liver (Figure 3-figure supplement 2J). To probe the cellular resource, we isolated hepatocytes and NPCs from SKO livers 2 days after PHx. Wnt10a was mainly produced by hepatocytes, whereas the major source of Shh might not be hepatocytes (Figure 3-figure supplement 2K). These results combined with the Shp2 rescue experiment suggest that the Wnt10a paracrine signal from SKO hepatocytes might be a local inducer of the colonies. Treatment of human HCC cells with Trametinib upregulated Wnt10a expression, and additional inhibition of PORCN suppressed CD133 induction (Figure 3-figure supplement 2L), suggesting a role of Wnt10a in CD133 induction under proliferative signal deficiency.

Interestingly, we observed CD133 on filament-like structures in colonized E-cadherin⁺ cells in primary SKO hepatocyte culture (Figure 5A). These CD133⁺ filament-like structures bridged neighboring cells in the colonies (Figures 5A and 5B), suggesting a direct cell-cell

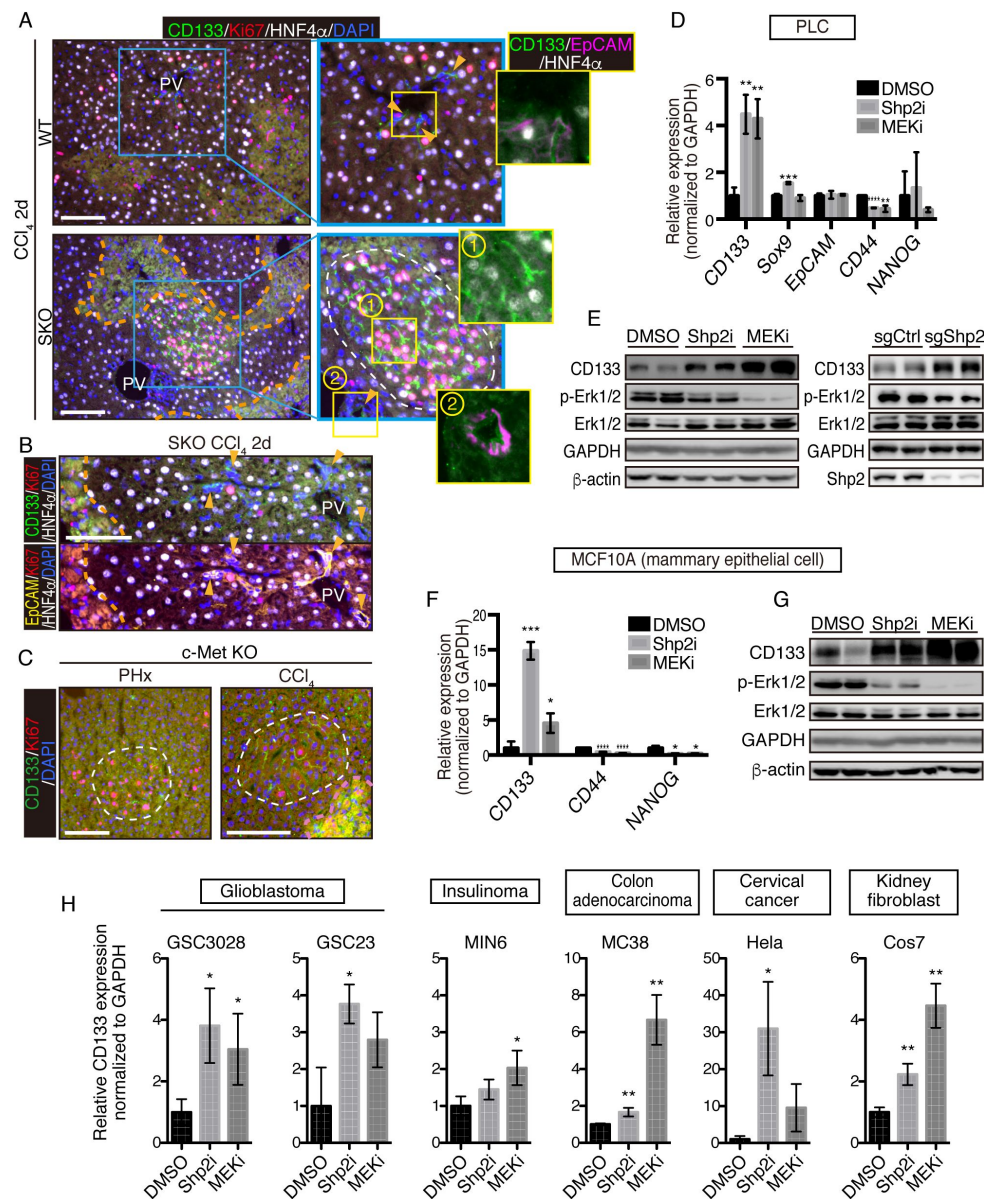


Figure 4.

CD133 induction during signal deficiency is a widely conserved mechanism

(A and B) Immunofluorescence of liver sections 2 days after CCl₄ injection. Proliferating hepatocytes were scattered in WT livers, whereas they were highly concentrated in CD133⁺ colonies (white dashed line) in SKO livers as shown in (A). CD133⁺/EpCAM⁺/HNF4α⁻ bile duct epithelial cells (arrowheads) were not associated with the colonies. Pink dashed lines, injured areas. PV, portal vein.

(C) Immunofluorescence of Met^{hep-/-} liver sections 2 days after PHx or CCl₄ injection. White dashed lines, CD133⁺ colonies. Pink dashed line, injured area.

(D) qRT-PCR analysis of PLC cell lysates treated with Shp2 or MEK inhibitors (Shp2i and MEKi). **P < 0.01, ***P < 0.001, ****P < 0.0001 (two-tailed unpaired t test, each compared with DMSO treatment). Means ± SD from 3 replicates are shown.

(E) Immunoblotting of PLC cell lysates treated with inhibitors or transfected with Shp2 targeting CRISPR vector. Guide RNA targeting the AAVS1 safe harbor site was used as a control (sgCtrl).

(F) qRT-PCR analysis of MCF10A cell lysates treated with the inhibitors. *P < 0.05, ***P < 0.001, ****P < 0.0001 (two-tailed unpaired t test, each compared with DMSO treatment). Means ± SD from 3 replicates are shown.

(G) Immunoblotting of MCF10A cell lysates treated with inhibitors.

(H) qRT-PCR analysis of lysates from various cell lines treated with Shp2 or MEK inhibitors.

*P < 0.05, ***P < 0.001, ****P < 0.0001 (two-tailed unpaired t test, each compared with DMSO treatment). Means ± SD from 3 replicates are shown.

Scale bars, 100 μm (A-C).

communication event. This direct communication model is consistent with the distinctive feature of clustered cells with extremely tight contacts *in vivo* and *in vitro* (**Figures 1B**, **1D**, **2A**, **3F** and **3H**). Given the filament-like staining pattern, we examined PLC cells with an antibody against human CD133. Indeed, we observed similar CD133⁺ filament structures inside and also connecting the cells (**Figure 5C**). Of note, the antibodies used for the mouse and human cells are different and do not cross-react with the other species. To further rule out a possible artifact in immunostaining for CD133, we expressed a Myc-tagged CD133 fusion protein in PLC cells and stained with antibodies against CD133 and the Myc tag. The two signals overlapped on the filament-like structure, confirming the subcellular localization of CD133 (**Figure 5-figure supplements 1A** and **1B**). As CD133 possesses five transmembrane domains and a membrane targeting signal peptide, we postulated that the observed filament-like CD133 staining represents CD133⁺ vesicles on filaments. Indeed, the CD133 signal co-localized with tubulin filaments (**Figure 5-figure supplement 1C**). CD133 overexpression led to a thicker pattern of CD133⁺ filaments, especially at high expression levels (**Figure 5-figure supplements 1A-1C**). Similar CD133⁺ structure was induced by Shp2 and MET inhibitors in MCF10A, HeLa and MC38 cells (**Figures 5D** and **Figure 5-figure supplement 1D**). Thus, these data suggest that CD133 is located on a trafficking machinery in proliferating cells.

CD133⁺ vesicles are induced by defective proliferative signaling

The filamentous staining profile of CD133 was different from the conventional droplet-like patterns often observed with endosomes or exosomes in a multi-vesicular body. To determine subcellular localization of CD133 and the CD133⁺ filaments, we performed stochastic optical reconstruction microscopy (STORM) imaging. Indeed, the CD133⁺ filaments observed at low resolution were CD133⁺ vesicles tightly associated with and distributed on tubulin filaments at super-resolution (**Figure 5E**). CD133 overexpression modified the morphology and topology of the vesicles (**Figure 5F**), suggesting a role of CD133 in the vesicle assembly and dynamics. Quantitative co-localization analysis confirmed physical location of CD133⁺ vesicles on tubulin filaments, while the value decreased after CD133 overexpression (**Figures 5E**, **5F**, and **Figure 5-figure supplement 1E**). The lowered co-localization value following CD133 overexpression likely reflected altered morphology of the vesicles, stretching away from the tubulin filaments, which also explained the bulky appearance in low resolution images (**Figure 5-figure supplements 1A-1C**). While this function is consistent with a reported role of CD133 on lipid membrane morphology/topology (Röper *et al.*, 2000; Thamm *et al.*, 2019), we observed CD133 on cell surface membrane only when exogenous CD133 was over-expressed (**Figure 5-figure supplements 1B** and **1F**). In agreement with STORM imaging, immunofluorescence and immunogold electron microscopy on a cryo-ultramicrotome section of the colonies in regenerating SKO liver further confirmed localization of CD133 on intracellular vesicles enriched between the apical lumens, rather than distributed on the apical membrane surface (**Figures 5G** and **Figure 5-figure supplement 1G**).

We isolated CD133⁺ vesicles from PLC cells using magnetic beads bound with anti CD133 antibody, with the yields dramatically increased after pre-treatment with Trametinib (**Figure 5H**). Visualized under EM, the sizes of CD133⁺ vesicles were estimated around 50 nm (**Figure 5I**), relative to EVs/exosomes that are usually at the sizes of 30-200 nm. Notably, CD133⁺ vesicles did not contain exosome markers CD9, CD63 or CD81, supporting their identity as a new type of vesicle, different from the classical exosomes (**Figure 5H**). Co staining endosome markers, RAB5A and RAB7, showed no overlap with CD133, with no CD133 signal detected in the multivesicular body where exosomes accumulated (**Figure 6-figure supplement 1A**). Evidently, CD133 was located on a filament network that connects neighboring cells, instead of endosomes or exosomes. Therefore, we give a name “intercellsome” to this new type of vesicle identified here.

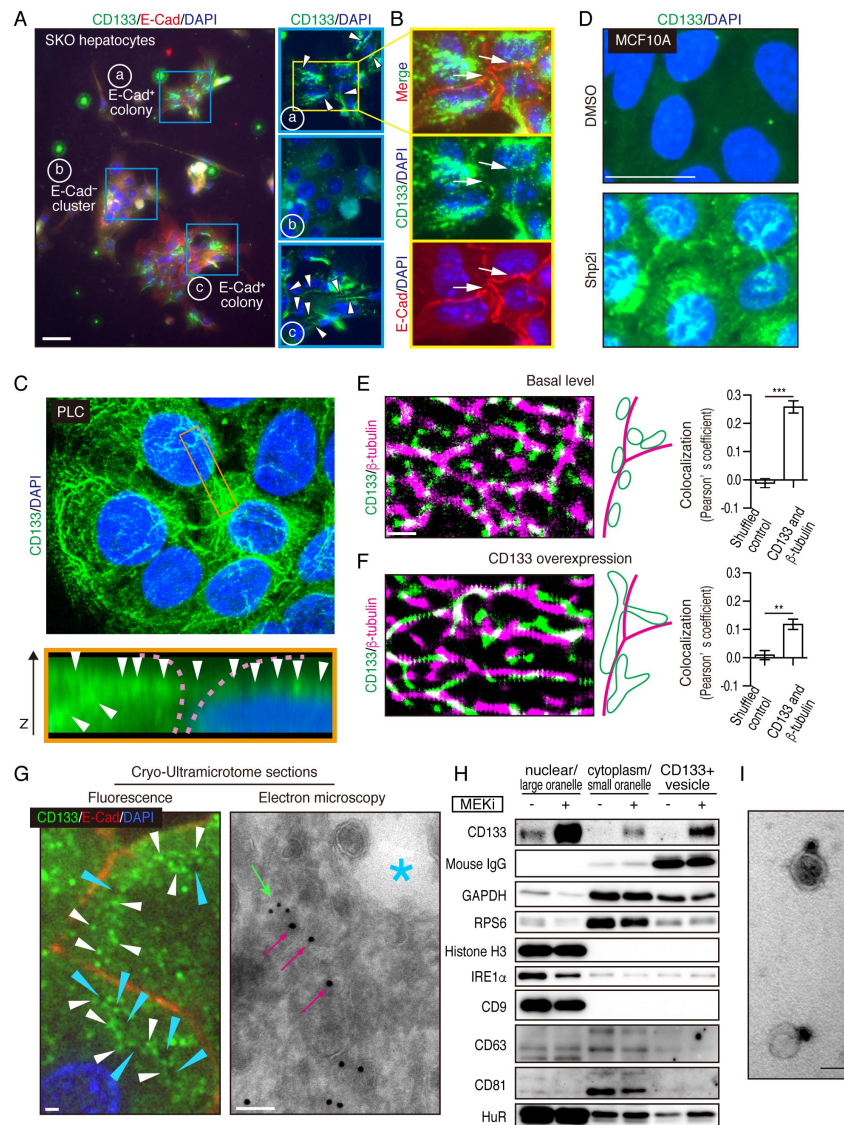


Figure 5.

Mitogenic signal deficiency induces CD133⁺ vesicles

(A and B) Immunofluorescence on in vitro colonies of primary hepatocytes from SKO liver.

CD133 was localized at filament-like structures in E-Cad⁺ colonies as shown by arrowheads in (A), which were connected between different hepatocytes as shown by arrows in (B).

(C) 3D-reconstituted confocal image of immunofluorescence on PLC cells. Lower panel shows the Z-plane section of the orange box area. Arrowheads indicate the CD133 signal on continuous filament like structures bridged between neighboring cells. Pink dashed lines indicate the cell surface.

(D) Immunofluorescence of MCF10A cells treated with Shp2 inhibitor.

(E and F) Super-resolution STORM images of immunofluorescence on PLC cells without (E) or with (F) CD133 overexpression. Colocalization of CD133 and β -tubulin was analyzed as Pearson's coefficient. Mismatched green and magenta channels from shuffled ROIs were measured as controls (Figure S4E). Means $\pm$ SEM from 6 images are shown. **P < 0.01, ***P < 0.001 (two-tailed unpaired t test).

(G) Immunofluorescence and Immuno-Gold EM images of cryo-ultramicrotome sections of SKO liver tissue after PHx. Cyan arrowheads and asterisk indicate apical lumens. White arrowheads indicate the CD133 signals aligned between apical lumens of neighboring cells. Light green arrow, CD133 staining (12 nm colloidal gold); Magenta arrows, α -tubulin staining (18 nm colloidal gold).

(H) Immunoblotting of CD133⁺ vesicles isolated from MEK inhibitor (MEKi)-treated PLC cells. Markers for different fractions were analyzed. CD133 antibody used for the vesicle isolation was IgG produced in mouse, which was detected by anti-mouse IgG antibody, showing efficient capture by the beads. Despite the efficient capture, the DMSO-treated PLC cells did not have much CD133⁺ vesicle to be bound with the antibody.

(I) EM image of Immunogold staining on the isolated vesicle.

Scale bars, 100 μ m (A), 25 μ m (D), 1 μ m (E, F, Fluorescence in G), 100 nm (EM in G), 50 nm (I).

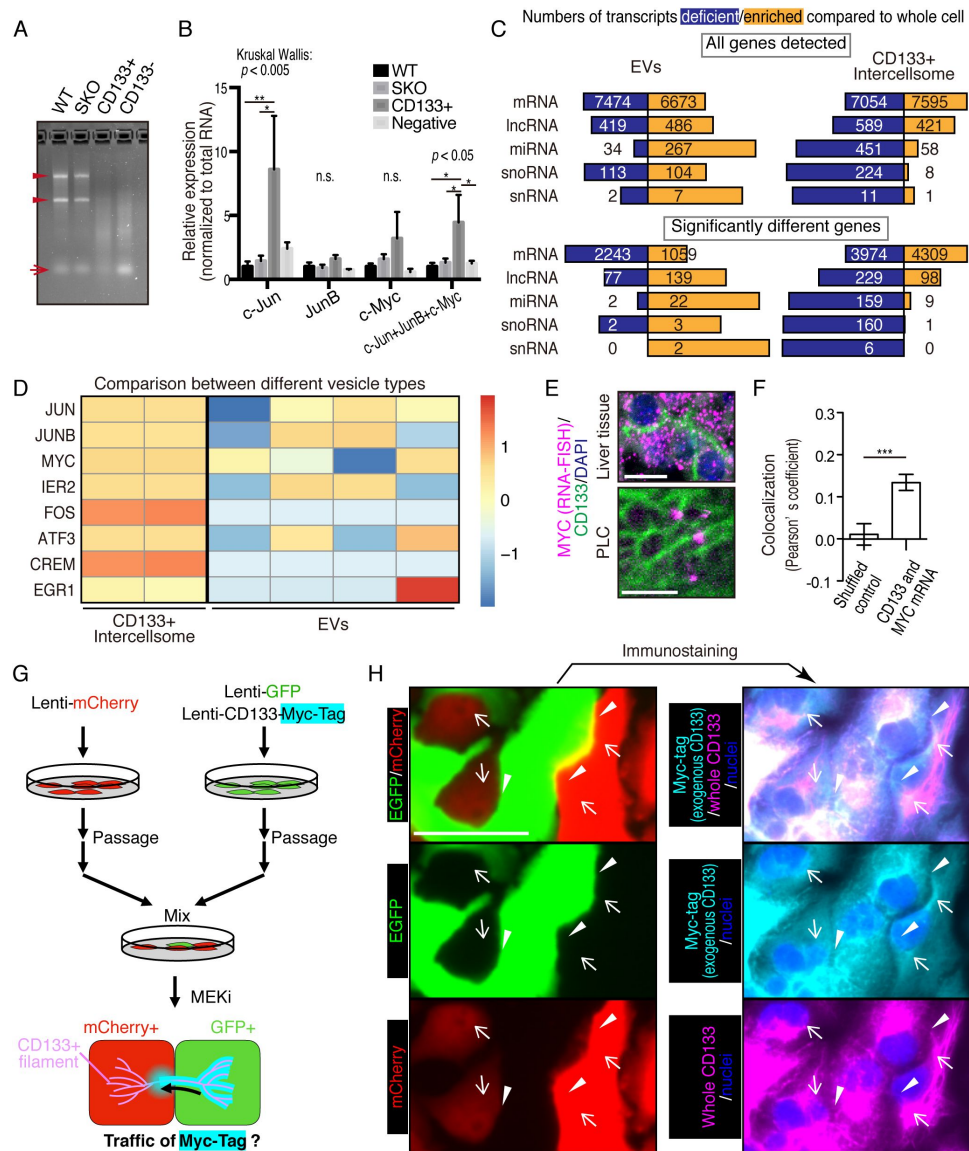


Figure 6.

CD133⁺ vesicles contain mitogenic mRNAs and traffic between neighbor cells

(A) Agarose gel electrophoresis of total RNAs extracted from WT and SKO livers and CD133-positive vesicle and negative fractions from SKO liver after PHx. Arrowheads show rRNAs and arrow shows microRNAs.

(B) qRT-PCR analysis of RNAs extracted from WT (3 mice) and SKO tissues (3 mice) and from CD133⁺ vesicles and CD133⁻ fractions (4 mice for both). Means $\pm$ SEM are shown. * $P < 0.05$, ** $P < 0.01$ (uncorrected Dunn's multiple comparison test, performed after Kruskal Wallis test). n.s., not significant.

(C) RNA-seq analysis of the different cell types and the whole cells. The bars indicate proportions between numbers of deficient and enriched gene transcripts in each RNA types.

(D) Comparison of IEG contents between the different vesicle types with RNA-seq. Different lanes indicate independent vesicle isolations.

(E) RNA-FISH for MYC mRNA and immunostaining for CD133.

(F) Quantitative colocalization analysis of CD133 and MYC mRNA in PLC cells shown in (E).

(E). Means $\pm$ SEM from 14 images are shown.

(G) Experimental design to detect the traffic of CD133⁺ vesicles between neighbor cells.

(H) Immunostaining of Myc-tag and CD133 on GFP⁺ and mCherry⁺ PLC cells mixed as shown in (G). Note that Myc-tag only indicates exogenous CD133, while CD133 indicates both endogenous and exogenous CD133. Myc-tag was primarily detected in the GFP⁺ cells, but also detected on the bridges (arrowheads) and in the mCherry⁺ cells (arrows). GFP was not detected at the same locations (arrowheads and arrows), indicating the specific traffic of CD133-Myc-tag.

Scale bars, 10 μ m (E) and 50 μ m (H)

CD133⁺ intercellsomes contain unique cargos

We isolated CD133⁺ vesicles from SKO liver lysates following PHx, and found that the vesicles were not enriched with signaling molecules, such as Stat3 and Erk1/2, compared to the CD133⁻ fractions (**Figure 6-figure supplement 1B** [↗](#)), suggesting that proliferative signaling proteins were not likely the most critical cargos. Interestingly, the CD133⁺ vesicles were markedly different from CD133⁻ fractions in the overall RNA profiling and contained minimal amount of rRNAs or micro-RNAs, with the majority being mRNAs (**Figure 6A** [↗](#)). The expression of immediate early-responsive genes (IEGs), including *c-jun*, *JunB*, and *c Myc*, were upregulated rapidly following PHx (Fausto, 2000 [↗](#); Haber *et al.*, 1993 [↗](#)). qRT-PCR analysis showed that these mitogenic mRNAs were loaded in CD133⁺ vesicles, with no enrichment detected for housekeeping gene transcripts, such as *Gapdh* and *Ppia*, (**Figures 6B** [↗](#) and **Figure 6-figure supplement 1C** [↗](#)), showing selective inclusion of mRNAs. Similar results were obtained with CD133⁺ vesicles isolated from PLC cells treated with Trametinib (**Figure 6-figure supplement 1D** [↗](#)).

We performed RNA-seq analysis of CD133⁺ vesicles isolated from Trametinib-treated PLC cells, extracellular vesicles (EVs) secreted by the same cells, and the whole cells. Isolation of the EV fraction was performed by the polymer-based precipitation method from the culture supernatant. Small RNAs, including miRNAs, snoRNAs and snRNAs, were less abundant in CD133⁺ vesicles than the whole cells (**Figure 6C** [↗](#)). In contrast, miRNAs were enriched in the EVs compared to the whole cells (**Figure 6C** [↗](#)). Although the majority of RNA species detected in EVs were still mRNAs, the data could be influenced by the library preparation with total RNAs, which missed lots of small RNAs. Nevertheless, the enrichment of miRNAs in the EVs was in sharp contrast to significantly enriched mRNAs in CD133⁺ vesicles (**Figure 6C** [↗](#)), clearly distinguishing the two types of vesicles. Compared to the whole cells, 4309 mRNAs were upregulated, with 3974 mRNAs downregulated, in CD133⁺ vesicles (**Figure 6C** [↗](#)). In particular, the RNA-seq detected enrichment of IEG transcripts in CD133⁺ vesicles, while most of these mRNAs were at low levels in the EVs (**Figures 6D** [↗](#) and **Figure 6-figure supplement 1E** [↗](#)). Consistently, single molecule in situ hybridization showed localization of *c-Myc* mRNA on CD133⁺ filaments, verifying the IEG existence in these vesicles (**Figures 6E** [↗](#) and **6F** [↗](#)).

To trace CD133⁺ vesicle activity, we established a GFP-labelled cell line that also expressed stably a CD133-Myc fusion protein. After mixing these cells with mCherry expressing cells without the fusion protein, we traced the Myc-tag signal (**Figure 6G** [↗](#)). Indeed, the CD133-Myc protein was transported from GFP⁺ cells to the neighboring mCherry⁺ cells (**Figure 6H** [↗](#)). Importantly, neither GFP nor mCherry signals migrated between the cells, suggesting selective CD133⁺ vesicle traffic intercellularly. We further established a cell line that stably expressed a CD133-GFP fusion protein and mixed with mCherry-expressing cells without the fusion protein. Correlative light and electron microscopy (CLEM) confirmed the traffic of CD133-GFP to the mCherry⁺ cells (**Figure 6-figure supplement 2** [↗](#)). Together, these results suggest a unique property of the vesicles, which contain particular cargos that are different from those in EVs.

mRNA sharing converts intercellular heterogeneity into intracellular diversity of IEGs

HuR is an RNA-binding protein that binds to AU-rich elements in mRNAs for their stabilization, and shuttles between nucleus and cytoplasm. This protein was loaded in CD133⁺ vesicles, as revealed by immunoblotting (**Figures 5H** [↗](#) and **7A** [↗](#)) and was localized to the apical side of CD133⁺ hepatocytes (**Figure 7-figure supplement 1A** [↗](#)). Besides the nuclei, immunostaining detected HuR in some peri-nuclear areas (**Figure 7B** [↗](#); arrows) and co localization with CD133 in PLC cells (**Figure 7B** [↗](#)). The CD133⁺ filaments bridged neighboring cells with HuR visible on the bridging filaments (**Figure 7B** [↗](#); arrowheads), suggesting migration of HuR-bound mRNAs directly from one cell to another. Treatment of PLC or MCF10a cells with Shp2 or Mek inhibitor markedly increased the density of HuR on CD133⁺ filaments (**Figures 7C** [↗](#) and **Figure 7-figure**

supplement 1B [↗](#)), with even brighter HuR signals than the nuclei. Given that CD133 was found to modify the vesicle patterns (**Figures 5E** [↗](#), **5F** [↗](#), **Figure 5-figure supplements 1A-1C** [↗](#)), we tested if CD133 overexpression also affected HuR trafficking. Indeed, the pattern of HuR changed together with altered CD133⁺ signals (**Figure 7D** [↗](#)). Lipofection of PLC cells with isolated CD133⁺ vesicles enhanced cyclin D1 expression, supporting a notion that mitogenic molecules were carried in the vesicles (**Figures 7E** [↗](#), **7F** [↗](#), **Figure 6-figure supplement 1D** [↗](#), **Figure 7-figure supplement 1C** [↗](#)). Colocalization of CD133 and HuR on the isolated vesicles was also demonstrated with an electron microscopy analysis (**Figure 7-figure supplement 1D** [↗](#))

We wondered how a potential exchange of mitogenic mRNAs within cell colonies could boost proliferative capacity. If the cells with impaired signaling had equally reduced amounts and types of mitogenic mRNAs, exchange would not be beneficial. We hypothesized that the expression of different IEGs was variably downregulated, which triggered intercellular exchange of mRNAs through CD133⁺ intercellsomes (**Figure 7G** [↗](#)). This process may convert intercellular heterogeneity to higher intracellular IEG diversity (calculated as entropy), without increasing total IEG expression levels. To test this theory, we performed scRNA-seq of WT and SKO hepatocytes at 4 hrs and 2 days following PHx, using our newly established protocol ([Chen *et al.*, 2021](#) [↗](#); [Liang *et al.*, 2022](#) [↗](#)). Hepatocyte identity was confirmed by unsupervised clustering and analyses of multiple markers, with contaminated NPCs excluded from further analyses (**Figure 7-figure supplement 2** [↗](#)). We focused attentions on IEGs, which were highly induced following PHx (Methods) and were detected in isolated CD133⁺ vesicles. Indeed, the heatmap demonstrated stochastic expression profiles of different IEG species in individual cells (**Figure 7-figure supplement 3A** [↗](#)). CD133⁺ cells were easily identified in SKO liver due to its high expression (**Figure 7-figure supplement 3B** [↗](#)). Of note, CD133⁺ cells were not enriched with particular hepatocyte subtype markers or stem cell markers, compared to CD133⁻ cells in SKO liver (**Figure 7-figure supplement 3C** [↗](#)). Principal component analysis (PCA) of the IEG expression profiles showed overlap of all samples with each other, instead of partitioning into distinct groups, suggesting no biased expression of specific IEGs in the CD133⁺ cell population in SKO liver (**Figure 7-figure supplement 3D** [↗](#)). Nonetheless, clustering of CD133⁺ cells in the center of PCA plot showed that despite the stochastic expression pattern, CD133⁺ cells were featured by relatively similar and stable levels of IEG species. By measuring the overall expression amounts of IEGs in the selected list, we found that the total IEG levels were higher in WT than SKO cells at 4 hr after PHx (**Figure 7H** [↗](#)). We next analyzed IEG diversity within individual cells and variations in the expression among cells. For fair comparison, the analysis was focused on Cyclin D1⁺ cells in the periportal area, to avoid influences of cell cycle and zonation in the liver (**Figure 7-figure supplement 3E** [↗](#), and Methods). Indeed, the intracellular IEG diversity was high and intercellular variations were low among CD133⁺ hepatocytes, relative to other groups (**Figures 7I** [↗](#) and **7J** [↗](#)). This data distinguished CD133⁺ cells of SKO liver from WT hepatocytes and also revealed that the increased entropy of IEGs was not due to restoration of a WT-like expression pattern (**Figure 7G** [↗](#)). A plot with SKO hepatocytes showed clearly higher IEG diversity in CD133⁺ than CD133⁻ cells, with similar total IEG expression levels (**Figure 7K** [↗](#)). Consistent with the significance of the selected IEGs for cell proliferation, we observed a positive correlation between Cyclin D1, IEG diversity and the total IEG levels (**Figure 7K** [↗](#)). When simulating mRNA exchanging, the plot pattern of CD133⁻ cells in SKO liver moved toward a striking resemblance to the CD133⁺ population, increasing IEG diversity within each cell (**Figures 7L** [↗](#) and **7M** [↗](#)). Mathematical modeling demonstrated that exchanging 1/12 of IEGs in a cell with 4 other cells twice was sufficient to reach the distinctive IEG profile of CD133⁺ cells (**Figures 7L** [↗](#), **7M** [↗](#), **Figure 7-figure supplements 4A-4C** [↗](#), and Methods). Based on the cellular and RNA-seq data, together with math modeling, we propose that reciprocal mRNA sharing within clustered CD133⁺ cells may drive an enhanced proliferative phenotype, despite signaling deficiency in the whole population.

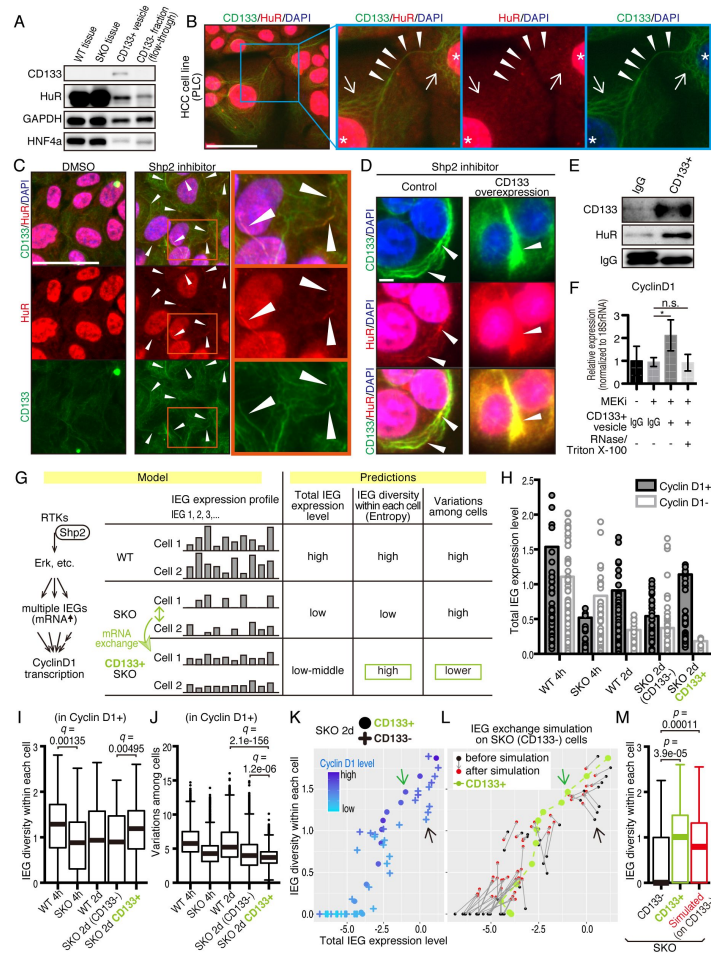


Figure 7.

CD133-mediated mRNA sharing converts intercellular heterogeneity into intracellular diversity

(A) Immunoblotting of HuR in the CD133⁺ vesicles from SKO liver after PHx. GAPDH and HNF4a were used as controls for cytoplasmic and nuclear fractions, respectively.

(B) Immunofluorescence on PLC cells. Arrows, peri-nuclear areas enriched with HuR in the cytoplasm; Arrowheads show colocalization of HuR on the CD133⁺ filament bridging two cells.

(C) Immunofluorescence on PLC cells treated with the Shp2 inhibitor (SHP099). Arrowheads show strong localization of HuR on the CD133⁺ filaments.

(D) Immunofluorescence on PLC cells transfected with CD133 expression vector treated with a Shp2 inhibitor (SHP099). Arrowheads show strong localization of HuR on the CD133⁺ filaments.

(E) Immunoblotting of CD133⁺ vesicles isolated from MEK inhibitor-treated PLC cells.

(F) qRT-PCR analysis of PLC cell lysates after treatment with CD133⁺ vesicles isolated from MEK inhibitor-treated PLC cells. RNase and Triton X-100 were used to digest the RNA content of the vesicles. *P < 0.05 (two-tailed unpaired t test). n.s., not statistically significant. Means ± SD from 3 replicates are shown.

(G) A model and predictions for single cell RNA-seq data analysis.

(H) Total IEG expression levels. Cyclin D1-positive and -negative cells were analyzed separately to evaluate the influence of cell cycle on the IEG analysis. See Methods section for what the bars and dots represent.

(I and J) Box plots (Tukey's) of IEG diversity within each cell (calculated as entropy) and IEG variations among cells. Analyses were focused on cyclin D1⁺ cells in all groups for fair comparison.

(K and L) Plot of intracellular IEG diversity against total IEG expression levels in SKO hepatocytes 2 days after PHx. Blue color gradient indicates cyclin D1 expression levels. For the simulation in (L), the parameters used were: Group of 5 cells, X=1/12, model 3 (see also Figure S8 and Methods). Green and Black arrows show typical profiles of CD133-positive and -negative cells, respectively.

(M) Box plot (Tukey's) of intracellular IEG diversity after simulation. The analysis was not limited to cyclin D1-positive or -negative cells. Note the simulation of the IEG exchange attracted the cells from cyclin D1-low profile to cyclin D1-high profile (K-M). Statistics was performed by Wilcoxon rank sum test adjusted by FDR in (I), (J) and (M).

Scale bars, 50 μm (B, C), 5 μm (D).

CD133 is necessary to sustain cell proliferation under signal deficit

Given the compensatory CD133 upregulation, we reasoned that a defective phenotype associated with CD133 loss could be amplified in a compound mutant also deficient for a pro-proliferative molecule. To test this theory, we crossed the *Prom1* KO mouse with a conditional *Shp2*^{flox/flox} mouse line, and deleted *Shp2* in hepatocytes by AAV-Cre virus infection. At 2 days after PHx, hepatocyte proliferation in E-Cad⁺ colonies was significantly suppressed in the *Prom1/Shp2* double KO (DKO) hepatocytes, compared to SKO control (**Figures 8A** and **8B**), although *Prom1* KO alone did not inhibit *Shp2*⁺ hepatocyte proliferation (**Figure 8B**). We also isolated hepatocytes from SKO and DKO (*Prom1* KO; *Shp2*^{flox/flox}; Alb-Cre) livers, and examined cell colonies formed in culture. Cell proliferation in E-Cad⁺ colonies was markedly lower in DKO than SKO hepatocytes (**Figures 8C** and **8D**). However, the basal low levels of proliferation were similar between SKO and DKO non colonized cells. We also observed higher proliferation rates of DKO hepatocytes within E Cad⁺ colonies than non-colonized cells in culture, suggesting that CD133 is unlikely the sole mediator of the intercellular communication event. Next, we labeled SKO hepatocytes by injecting AAV-GFP into SKO mice, and mixed the GFP-labelled SKO cells with DKO hepatocytes at 1:10 ratio in culture (**Figure 8E**). If CD133 had a cell-autonomous function, the SKO cells in a DKO cell pool were expected to proliferate similarly as in a pure SKO cell pool. For example, if CD133 serves as a receptor, SKO cells in the mixture should still receive a ligand signal from DKO cells. By contrast, the SKO cells scattered within DKO colonies only proliferated as much as DKO cells (**Figures 8F** and **8G**), arguing against a cell autonomous role of CD133. CD133 removal in DKO cells did not abolish direct cell-cell contact, with tight cell adhesions maintained, as shown by E-cad staining (**Figure 8F**).

Given the abundant CD133 expression in intestinal crypt, we isolated crypt cells from WT and *Prom1* KO mice for organoid culture in vitro. Consistent with previous data (Barker *et al*, 2010), CD133 was expressed in intestinal crypt cells, likely including both the Lgr5⁺ stem cells and the progenitor-like transit amplifying cells (**Figure 8-figure supplement 1**). Consistent to previous data, no significant difference in cell proliferation was observed between WT and *Prom1* KO organoids (**Figures 8H** and **8I**). However, treatment with Trametinib exhibited more prominent inhibition on cell proliferation in the crypt buds of *Prom1* KO organoids than the WT control (**Figures 8H** and **8I**), indicating higher sensitivity of CD133-deficient intestinal cells to impaired Ras-Erk signaling. Together, these data illustrate a functional role of CD133 in various cell types for strive to proliferate under signal deficit.

Discussion

Originally aimed at deciphering hepatocyte proliferation in liver regeneration, this study has unveiled a cell-cell communication mechanism shared in various cell types under defective proliferative signaling. Interestingly, this was deciphered through expression of CD133, a functionally mysterious molecular marker for stem cells and cancer stem cells. By taking multidisciplinary approaches, especially super-resolution microscopy and scRNA-seq, we obtained data that suggest a role of CD133⁺ vesicle for intercellular exchange of mitogenic mRNAs. Although CD133 was thought to label stem cells in normal and tumor tissues, its expression is not restricted to the stem cell populations in either case. We found that CD133 expression was transiently induced in *Shp2*- and *MET*-deficient hepatocytes in regenerating livers damaged by PHx or CCl₄. Of note, the CD133 upregulation was not accompanied by increase in other stem/progenitor cell markers or associated with cell lineage transition. Further, reversible CD133 induction was detected in several types of cancer cells following suppression of RTK-Ras-Erk signaling by chemical inhibition of *Shp2* or *Mek*. These data on dynamic CD133 expression suggest that its expression pattern is not strictly associated with cell identity but rather is modulated by intra- and inter-cellular signals, which may clarify some conflicting data and notions documented in the

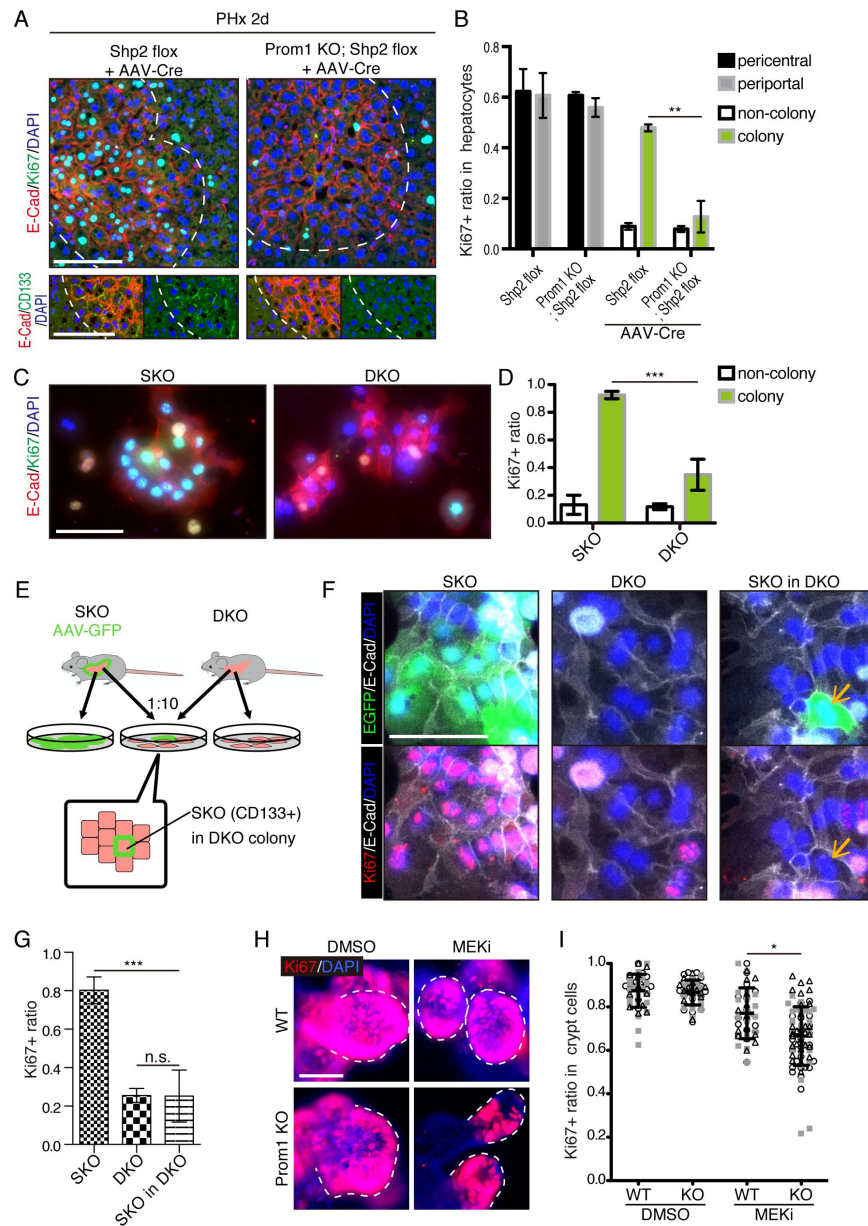


Figure 8.

CD133 is required for cell proliferation under proliferative signal deficit

(A and B) Immunofluorescence (A) on liver sections of WT and Prom1 KO mice 2 days after PHx with or without Shp2 deletion using AAV-Cre and quantification of Ki67⁺ ratio in hepatocytes in the indicated genotypes (B). Separate analyses of pericentral and periportal hepatocytes showed insignificance of zonal difference. $^{**}P < 0.01$, (two-tailed unpaired t test). Means $\pm$ SEM are shown. $n = 3, 3, 4$ and 5 mice, respectively.

(C and D) Immunofluorescence (C) on primary hepatocytes isolated from SKO and Shp2/Prom1 double KO (DKO) mouse livers and quantification of Ki67⁺ ratio (D). Images of representative colonies are shown. $^{***}P < 0.001$, (two-tailed unpaired t test). Means $\pm$ SD from 4 wells are shown.

(E) Experimental design with primary hepatocytes isolated from GFP labeled SKO liver and unlabeled DKO liver. E-Cad⁺ colonies were analyzed.

(F) Immunofluorescence images of E-Cad⁺ colonies in SKO, DKO and mixed culture as shown in (E). The arrows show a GFP⁺ SKO cell forming a part of the colony with the surrounding DKO cells.

(G) Quantification of the Ki67 ratio in E-Cad⁺ colony-forming cells shown in (F). $^{***}P < 0.001$ (two-tailed unpaired t test). n.s., not statistically significant. Means $\pm$ SD from 3 wells are shown.

(H and I) Immunofluorescence (H) on WT and Prom1 KO mouse intestinal organoids treated with MEK inhibitor (MEKi) and quantification of Ki67⁺ ratio in the crypt cells (I). Dashed lines indicate the crypt buds. $^{*}P < 0.05$, (two-tailed unpaired t test). Means $\pm$ SD from 3 wells are shown. Each dot represents each crypt buds, and symbols indicate each well.

Scale bars, 100 μ m (A, C and F) and 50 μ m (H).

literature. Given that proliferative signaling can be frequently disturbed in health and diseases, upregulated CD133 expression and function is likely a conserved mechanism for cell proliferation under stress in various cell types.

This study identified CD133-associated filaments that connect neighboring cells (**Figure 5**). Super-resolution microscopy and immuno-EM further demonstrated that CD133 was primarily located on intracellular vesicles. The CD133⁺ vesicles may mediate intercellular sharing of materials between tightly contacting cells with minimal diffusion. This putative role of CD133 explains why CD133/Prom1 deletion did not affect general development in mice, as CD133⁺ vesicles are mainly generated in stress responses. However, CD133 is unlikely the sole molecule responsible for formation of the vesicles; other molecules may also participate in this important cellular event. It is also plausible that cells use a CD133-independent mechanism to cope with stress in the absence of CD133. Further, this study did not exclude a possibility that CD133 has other functions in cellular activities. Although we identified intracellular CD133⁺ vesicles here, this molecule with five transmembrane domains may play a different role on cell surface in a context that was not studied here. CD133⁺ vesicles were reportedly detected in internal and external body fluids (Marzesco, 2013), although their cell origin and functional significance are unclear. It is conceivable that these reported CD133⁺ vesicles were released from damaged cells rather than excreted by healthy cells.

Unlike exosomes or EVs (Valadi *et al.*, 2007), CD133⁺ vesicles are enriched with mRNAs rather than miRNAs, especially the IEG transcripts. Math modeling based on the scRNA-seq data suggests that exchanging mRNAs between CD133⁺ cells could convert intercellular stochastic heterogeneity into higher intracellular diversity (entropy). Simulating the process with the math model reveals that reciprocal mRNA exchange among defective cells is sufficient to restore a proliferative capacity of CD133⁺ SKO hepatocytes deficient for proliferative signaling. Further experimental evidence is required to validate or modify this intercellular mRNA exchange theory. The diversity of mitogenic mRNAs is likely more important than upregulation of individual genes during cell proliferation. In this case, while the sharing of mRNAs to increase the entropy is used by non-stem cells for compensation of signal deficiency, stem cells may more often require this mechanism to maintain their functions, especially under stress. Indeed, recent scRNA-seq analyses suggest intracellular transcriptional diversity (entropy) as a common feature of stem cells (Grün *et al.*, 2016; Gulati *et al.*, 2020; Teschendorff & Enver, 2017). An intracellular transcriptional diversity can be calculated as Shannon's entropy or a count of gene transcript species within individual cells. Due to higher entropy of mRNAs, stem cells can shift their identity via differentiation or maturation, during which the transcriptional diversity is lowered for specific cell lineages. This property may explain why stem cells are more resilient in general, as higher entropy makes a more robust buffering capacity against various disturbances. Therefore, the intracellular transcriptional diversity is not merely a signature of stem cells but is functionally required for such robustness. Despite common features of various stem cells, there is no universal molecules that can identify or define all types of stem cells. Massive genomic data analyses in the literature have revealed different sets of biomarkers expressed in various stem cells. However, the intracellular diversity is likely a common property of all stem cells as well as actively dividing non-stem cells, which is maintained at least in part by CD133⁺ vesicles.

Despite the controversial concept of CSC and CD133 as its biomarker, many reports have documented a significant correlation between CD133 expression, drug resistance and prognosis of cancer patients. The CD133⁺ cancer cells are reportedly more resistant to radio/chemotherapies and thus likely responsible for tumor recurrence (Bao *et al.*, 2006; Bertolini *et al.*, 2009; Ma *et al.*, 2008; Sarvi *et al.*, 2014). We observed markedly increased expression of CD133 in several tumor cell lines following treatment with growth-inhibitory compounds, which are already used in clinical treatment or trials (**Figure 4**). These data suggest plasticity of CD133 expression in acute response of tumor cells to anti-tumor drugs, and a group of tumor cells may manage to proliferate through intercellular communications upon inhibition of proliferative signals. This

swift response to drug treatment does not require new mutations and is unrelated to the CSC identity. If considering “stemness” as a trait rather than an identity, it is likely that any cancer cells acquiring high transcriptional diversity may show some signs of stemness, such as drug resistance. The CD133 expression and the CD133⁺ vesicles are induced transiently in various cancer cell types, independent of their differentiation status. Thus, this study may have disclosed a long-sought CD133 function in cancer, further characterization of which may clarify the controversial issues regarding CD133 and the so-called cancer stem cells.

Many questions remain to be answered for this new type of vesicles, including their biogenesis, structure, components (RNAs, proteins and lipids) and functions. Although our data suggest direct traffic of CD133⁺ vesicles between adjoining cells, it is unclear how these vesicles migrate from one cell to another, going through tunneling nanotube-like structures or through the protrusions at donor and recipient cell surface via budding and endocytosis. As they were not localized to multivesicular bodies, CD133⁺ intercellsomes are unlikely originated from the same pathway as the exosomes. The exact function and mechanism of CD133 in intercellsomes are yet to be elucidated, although our data are in agreement with previous results showing association of CD133 with cholesterol and lipid membrane morphology changes (Röper *et al.*, 2000 [DOI](#); Thamm *et al.*, 2019 [DOI](#)). Answering these questions will likely elucidate a new intercellular communication mechanism associated with stress responses, under physiological and pathological conditions. Simultaneously suppressing intracellular proliferative signaling and disrupting the compensatory intercellsome function may be a new strategy to overcome drug resistance and tumor relapse.

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Competing interests

The authors declare no competing interests.

Materials and Methods

Animals

Hepatocyte-specific Shp2 knockout (*Alb-Cre; Shp2^{flox/flox}*) or SKO mice were generated as described previously (Bard-Chapeau *et al.*, 2011). c-Met^{flox/flox} mice (FVB;129P2-*Met^{tm1Sst}/J*) were purchased from the Jackson Laboratory and bred with Alb-Cre mice to produce hepatocyte specific c-Met KO mice. Prom1 KO mice (B6N;129S Prom1^{tm1(cre/ERT2)Gib/J) were purchased from the Jackson Laboratory and bred with Shp2^{flox/flox} and SKO mice. Shp2^{flox/flox} littermates were used as WT control mice. All experimental procedures were approved by the UCSD Institutional Animal Care and Use Committee (IACUC). PHx was performed on mice at age of 8-10 weeks with the use of Buprenorphine hydrochloride (Par Pharmaceutical) as a pain relief for the surgery. CCl₄ (20% in 100 µl corn oil/20g of body weight; Sigma) was intraperitoneally injected into 8-10 week-old mice.}

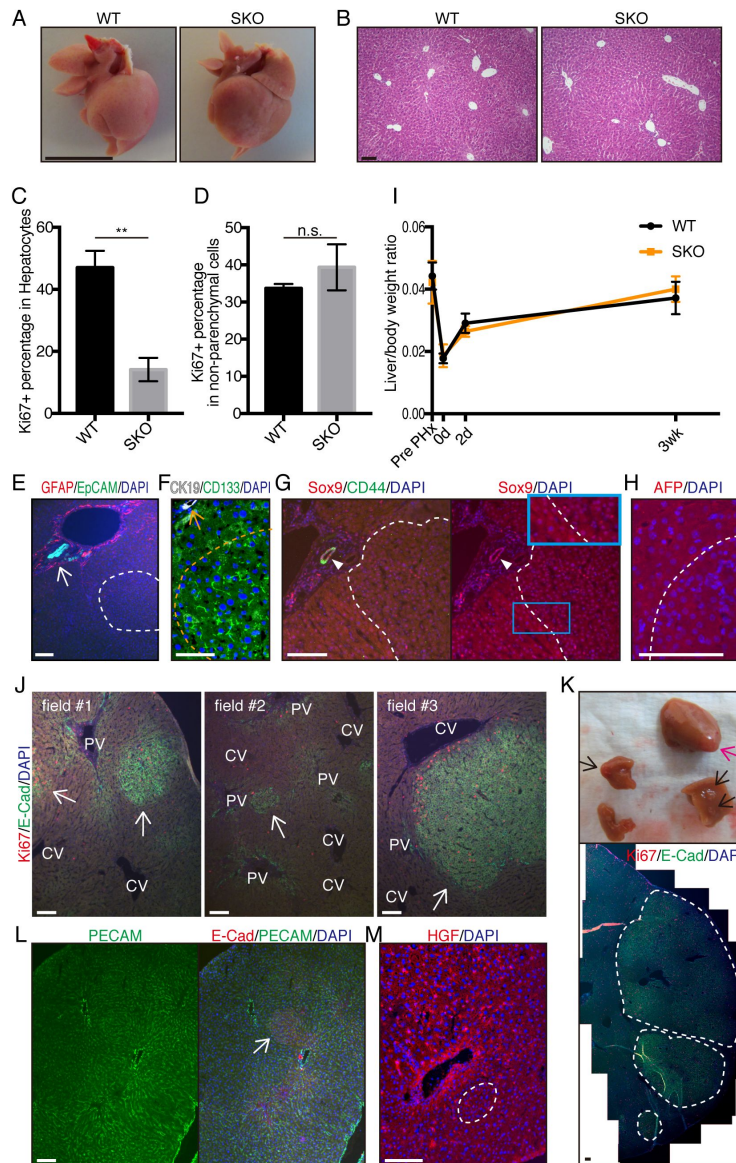


Figure 1-figure supplement 1.

Liver regeneration of WT and SKO livers after PHx

- (A) General morphology of livers 2 days after PHx.
- (B) H&E staining of liver sections 2 days after PHx.
- (C and D) Ki67⁺ percentages in HNF4α⁺ hepatocytes (C) and HNF4α⁻ NPCs (D) 2 days after PHx. Means ± SEM are shown, n=3 per group, **P < 0.01 (two-tailed unpaired t test). n.s., not significant.
- (E) EpCAM was highly expressed in biliary epithelial cells (arrow), but not in the colony (dashed line). Stellate cells (GFAP) were not altered around the colony.
- (F) CK19 was highly expressed in biliary epithelial cells (arrow), but not in the colony (dashed line). CD133 was highly expressed in the colony.
- (G) Sox9 and CD44 were highly expressed in biliary epithelial cells (arrowhead), but was not upregulated in the colony (dashed line) compared to surrounding hepatocytes.
- (H) AFP showed no difference between the colony (dashed line) and the surrounding tissue.
- (I) Liver/body weight ratios after PHx. Means ± SEM from 3 or more mice analyzed for each time point and group are shown.
- (J and K) Immunofluorescent staining of SKO liver sections 3 weeks after PHx. Colonies were at various sizes and locations as shown by arrows in (J). Macroscopic colonies were found as shown by arrows and dashed lines in (K). Immunofluorescent image in (K) corresponds to the magenta arrow in the liver image. PV, portal vein; CV, central vein.
- (L) Vasculature shown by PECAM did not distinguish the colony (arrow).
- (M) HGF expressed by NPCs was not concentrated in the colony (dashed line). Scale bars, 1 cm (A) 100 μm (B, E-G, I-L).

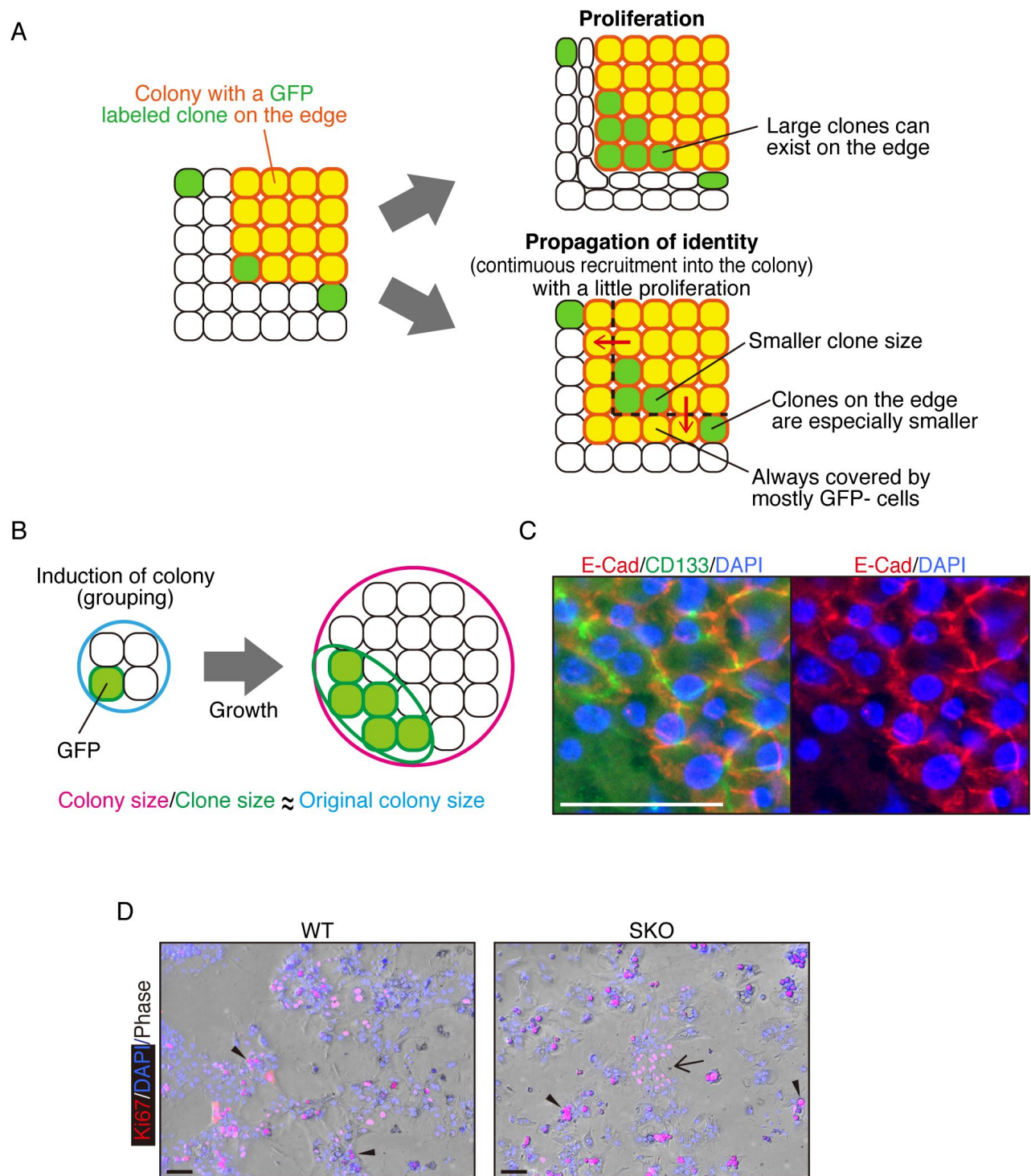


Figure 3-figure supplement 1.

Clonal analysis of hepatocytes

(A) Predicted outcomes of different dynamics.

(B) Colony size/clone size ratios can be used to roughly estimate the original colony sizes.

(C) Immunofluorescence of SKO liver sections 2 days after PHx, showing mixture of binuclear and mononuclear hepatocytes in the colony.

(D) Immunofluorescence of primary hepatocyte cultures from WT and SKO mice. The arrow indicates an in vitro colony in Shp2-deficient hepatocyte culture. Arrowheads, dying cells with high autofluorescence.

Scale bars, 50 μ m (C) and 100 μ m (D).

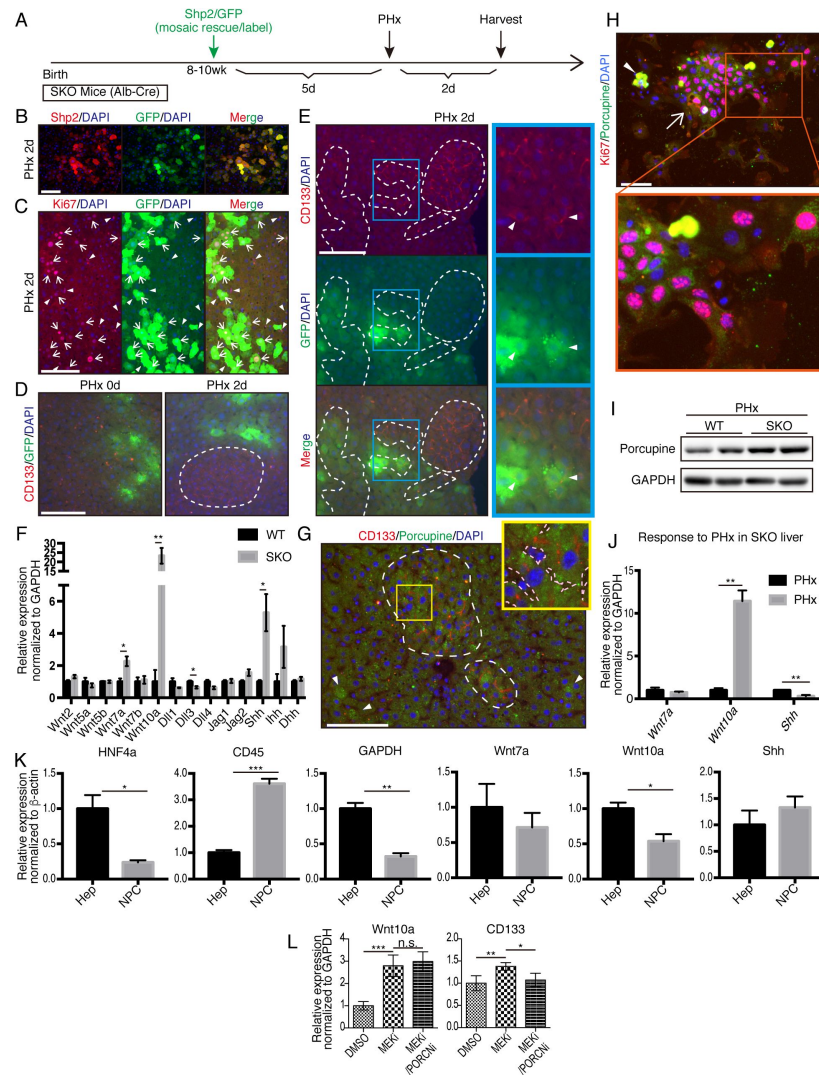


Figure 3-figure supplement 2.

Molecular analysis of intercellular signals during colony induction

(A) Schematic illustration of mosaic rescue experiment of Shp2 in Shp2-deficient liver.

(B) Immunofluorescence showing Shp2-rescued hepatocytes also labeled with GFP.

(C) GFP⁺ hepatocytes were proliferating (Ki67⁺; arrows), while other proliferating cells were not hepatocytes (arrowheads). Areas with colonies were not included here.

(D) Colonies were frequently constituted by GFP-negative hepatocytes (dashed line). Orange dots are non-specific stains of debris caused by HTVi.

(E) Some GFP⁺ hepatocytes showed expression of CD133 (arrowheads). Dashed lines show CD133-positive areas.

(F) qRT-PCR analysis of WT and SKO liver lysates 2 days after PHx. n=3, means ± SEM are shown, *P < 0.05, **P < 0.01 (two-tailed unpaired t test).

(G) Immunofluorescence on SKO liver section 2 days after PHx. White dashed lines mark Porcupine⁺ cells enriched in CD133⁺ colonies, rarely detected in non-colony area (arrowheads). Pink dashed lines: vasculature.

(H) Immunofluorescence on the colonies (arrows) in Shp2-deficient hepatocyte culture in vitro. Arrowhead, dying cells with high autofluorescence.

(I) Immunoblot analysis of WT and SKO liver lysates 2 days after PHx.

(J) qRT-PCR analysis of SKO liver tissue lysates without PHx (0d, 3 mice) and with PHx (2d, 3 mice). Means ± SEM are shown. **P < 0.01 (two-tailed unpaired t test).

(K) qRT-PCR analysis of hepatocyte and NPC fractions from SKO livers (3 mice) 2 days after PHx. HNF4a and CD45 were used as positive controls for the fractionation. β-actin was used for normalization, because GAPDH was highly expressed by hepatocytes. Means ± SEM are shown. *P < 0.05, **P < 0.01, ***P < 0.001 (two-tailed unpaired t test).

(L) qPCR analyses of PLC cells treated with Mek inhibitor and Porcn inhibitor. Means ± SD from 4 wells are shown. *P < 0.05, **P < 0.01, (two-tailed unpaired t test). n.s., not significant.

Scale bars, 100 μm (B-E, G and H).

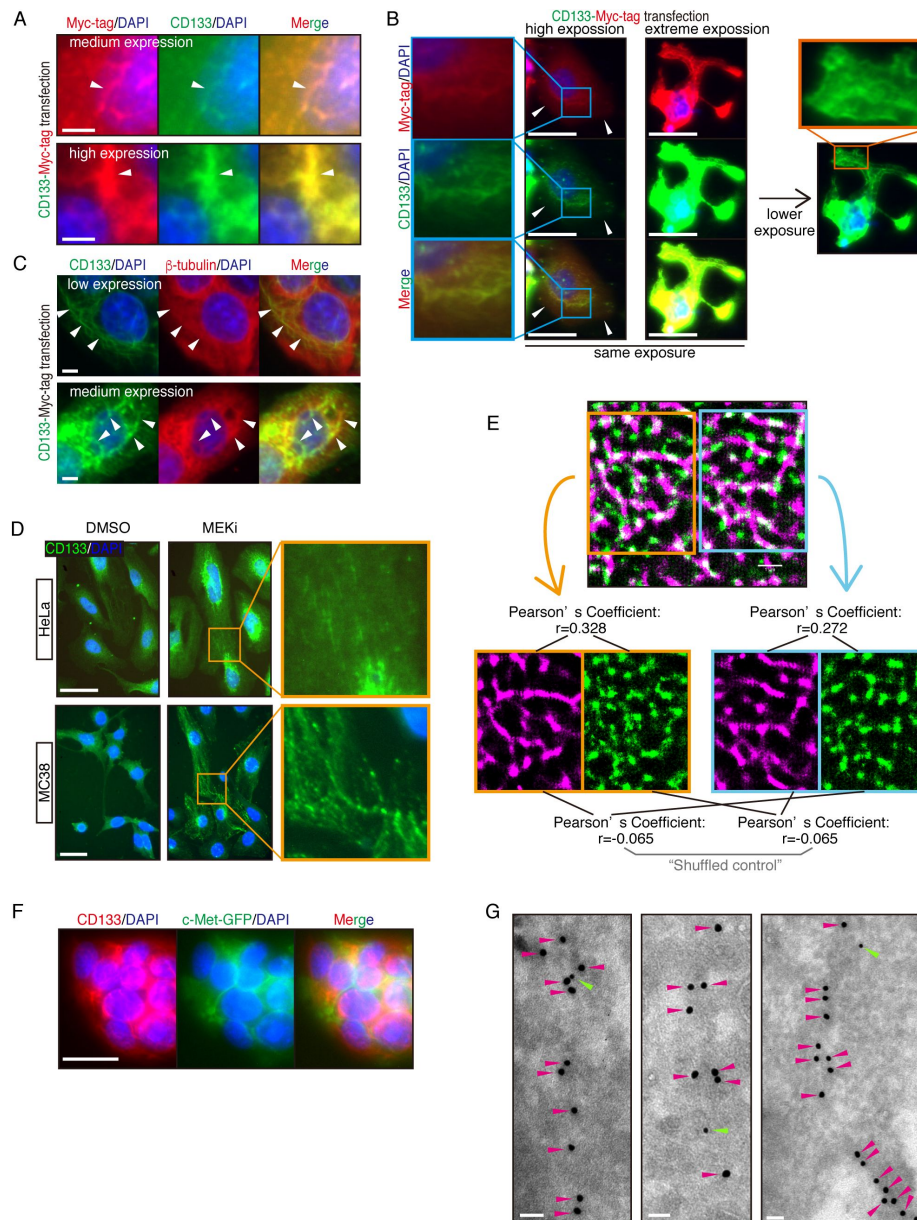


Figure 5-figure supplement 1.

Analyses of CD133 localization

(A-C) Immunofluorescence on PLC cells transfected with CD133-Myc-tag fusion protein expression construct. Cells with different expression levels are shown as representatives. In (A) and (C), exposure times for the CD133 signals were adjusted separately to clearly demonstrate the patterns, rather than intensities. Note that cells with higher CD133 expression displayed bulky pattern of the filaments, with maintained colocalization with tubulin filaments. Arrowheads, CD133⁺ filaments. As shown in (B), the overexpressed CD133 primarily localizes to the filaments, without detectable membrane localization (arrowheads). With extreme overexpression, CD133 can also localize to the cellular membrane surface, altering the morphology of the surface (right panels).

(D) Immunofluorescence on HeLa and MC38 cells.

(E) Colocalization of CD133 and β -tubulin was analyzed as Pearson's coefficient. Mismatched green and magenta channels from shuffled ROIs were measured as controls.

(F) Immunofluorescence on PLC cells transfected with cMet-GFP. cMet, an HGF receptor, shows the cell surface. CD133 did not colocalize with cMet-GFP on the cell surface, but instead localized to the filaments.

(G) Immuno-Gold EM images of cryo-ultramicrotome sections of SKO liver tissue after PHx. Light green arrowheads, CD133 staining (12nm colloidal gold); Magenta arrowheads, α tubulin staining (18nm colloidal gold).

Scale bars, 5 μ m (A and C), 50 μ m (B and D), 25 μ m (F) and 50nm (G).

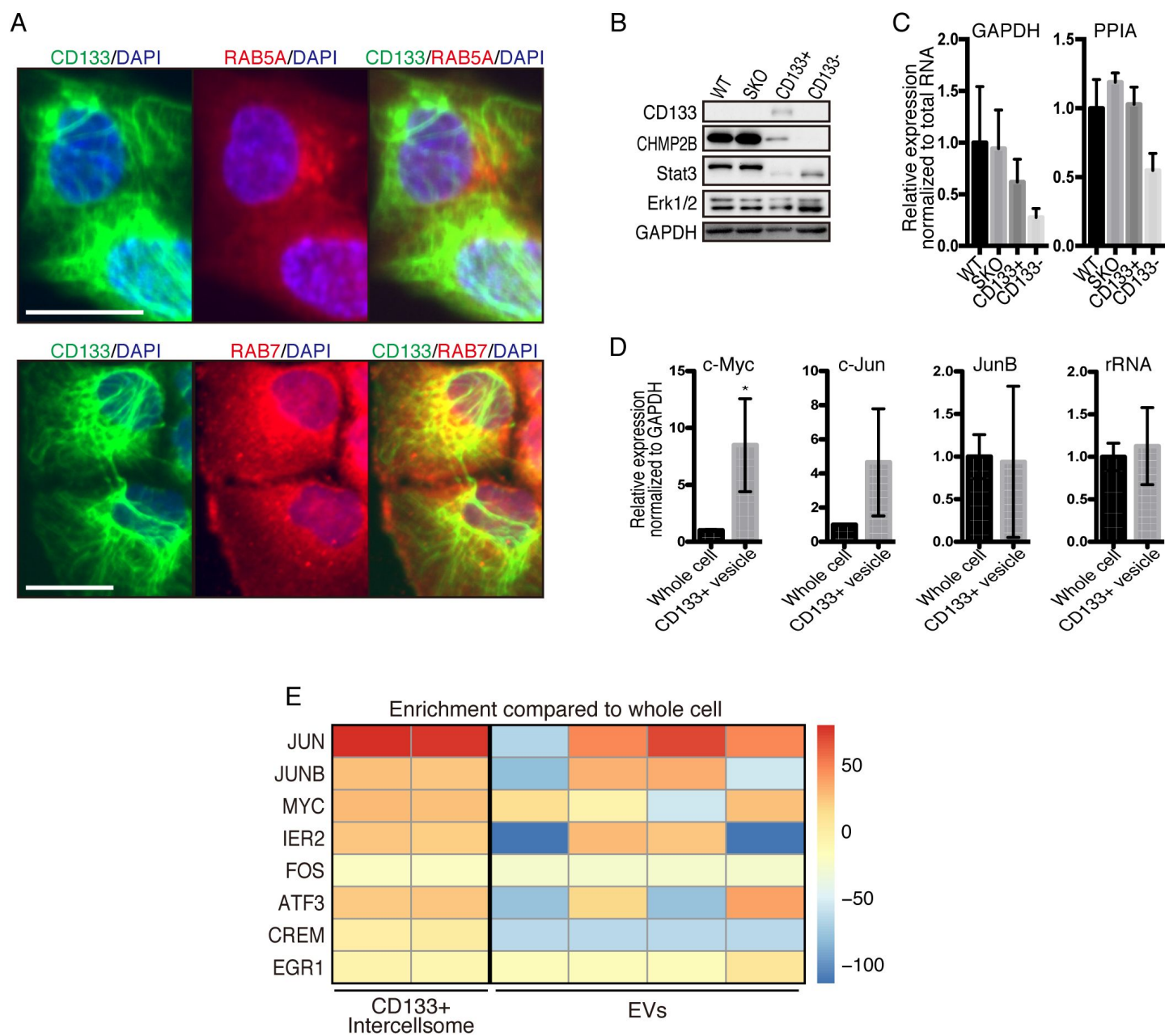


Figure 6–figure supplement 1.

Analyses of CD133⁺ vesicles

(A) Immunofluorescence for endosomal markers on PLC cells.

(B) Immunoblotting of CD133⁺ vesicles isolated from SKO liver after PHx. CHMP2B is a protein that is generally involved in vesicle formation.

(C) qRT-PCR analysis of housekeeping genes in RNAs extracted from WT (3 mice) and SKO tissues (3 mice) and from CD133⁺ vesicles and CD133⁻ fractions (4 mice for both). Means $\pm$ SEM are shown.

(D) qRT-PCR analysis of RNAs extracted from CD133⁺ vesicles from PLC cells treated with MEK inhibitor.

Scale bar, 25µm (A).

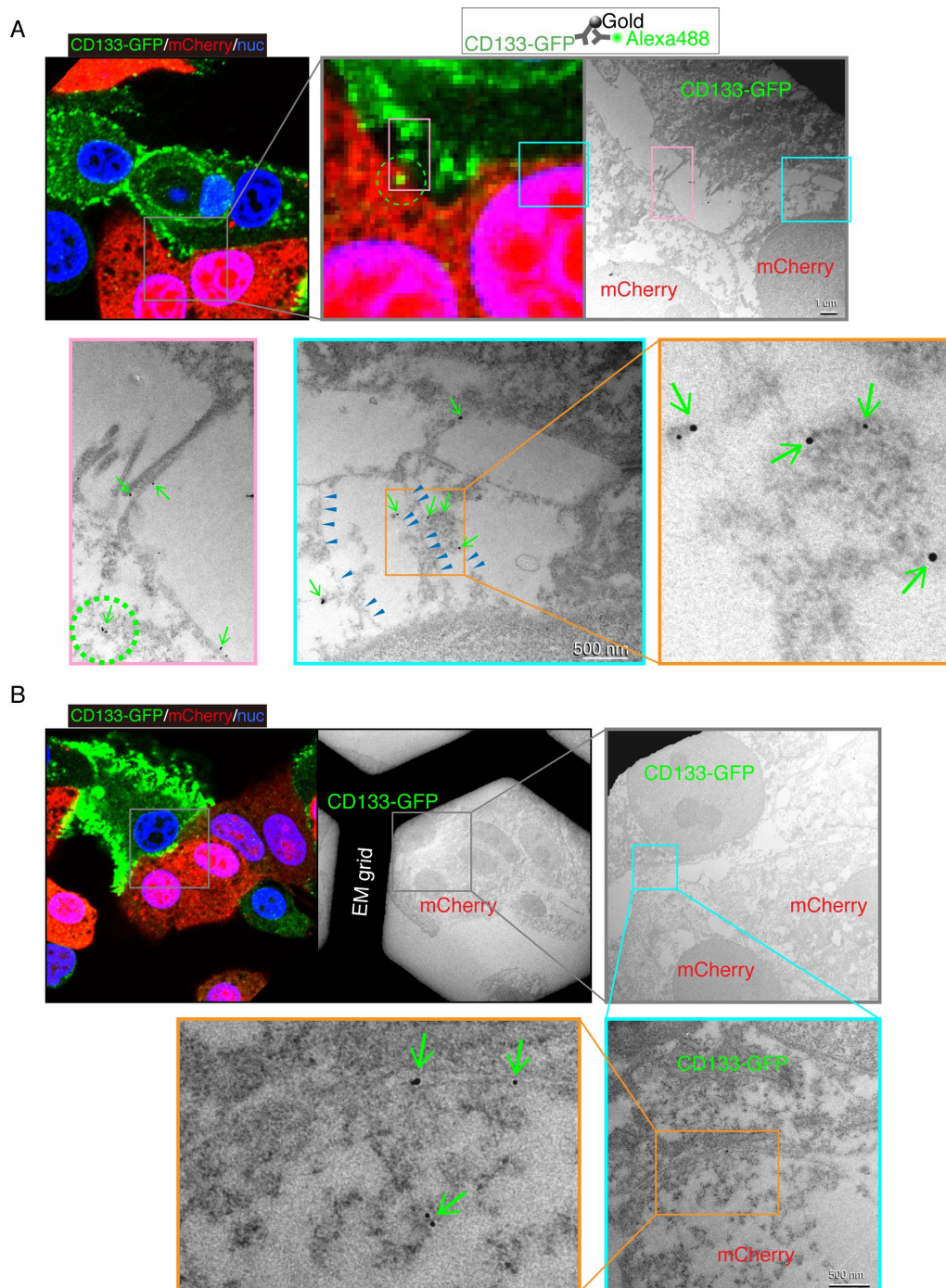


Figure 6-figure supplement 2.

Intercellular traffic of CD133⁺ vesicles

(A) Correlative light and electron microscopy (CLEM) analysis of the intercellular traffic demonstrated in **Figures 6G-H**. Instead of co-transfection of lenti-GFP and lenti-CD133-Myc as in **Figures 6G** and **H**, lenti-CD133-GFP was used, and the CD133-GFP expressing cells were mixed with mCherry-expressing cells. CD133-GFP was double-labeled with nano-gold, which was enhanced by silver and fluorophore Alexa488. The gold/silver signals vary in sizes due to variable silver enhancement. The gold/silver signals were detected around the boundaries between the CD133-GFP⁺ donor cells and the mCherry⁺ recipient cells, indicating intercellular traffic of the exogenous CD133-GFP protein (arrows). Some filament-like structures were also observed (arrowheads). The dashed circles show a corresponding signal in the fluorescent image and EM image.

(B) Another representative image in the same experiment as in (A). The gold/silver signals were observed between the donor and recipients as well as inside the recipient cell (arrows).

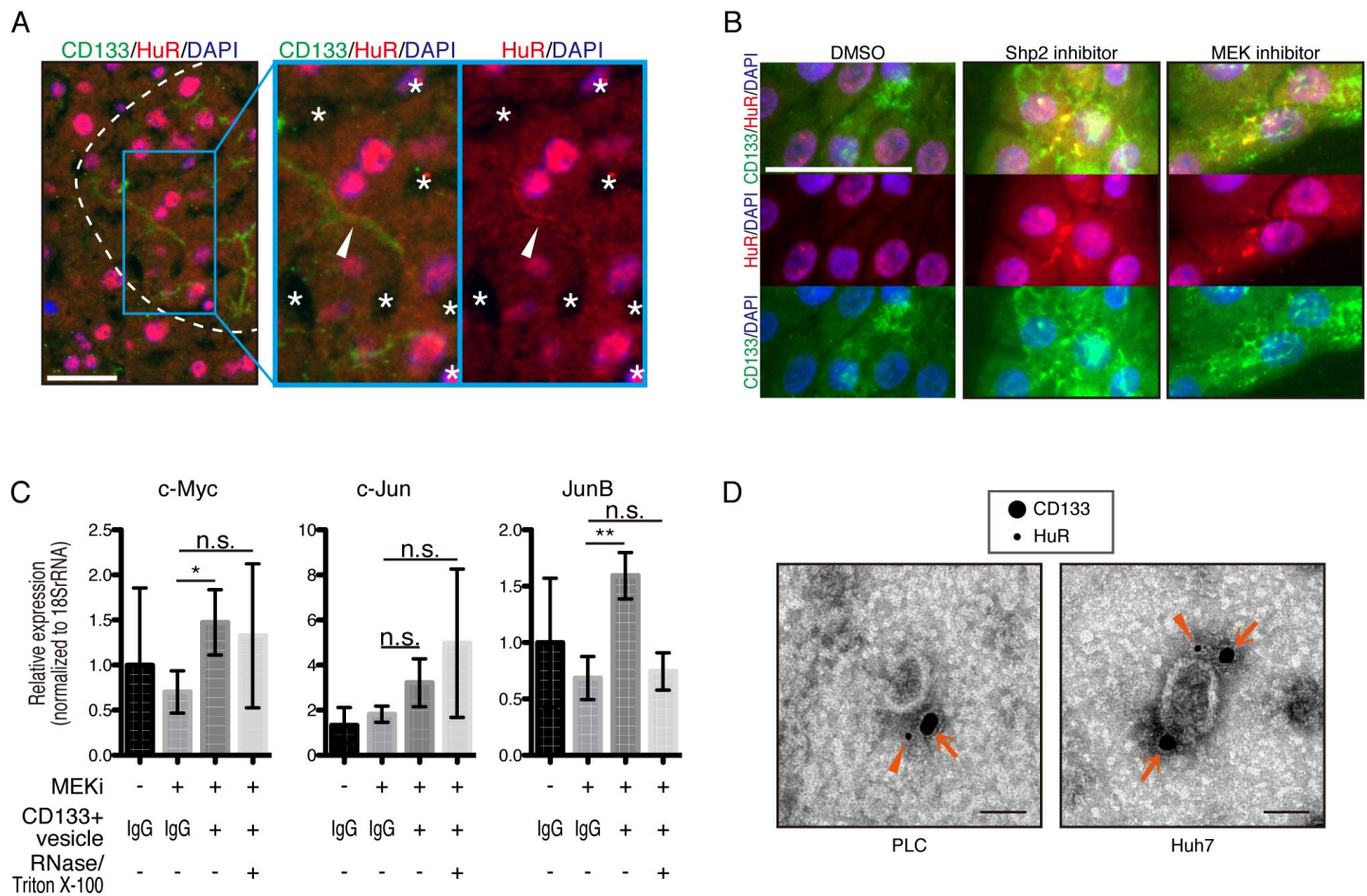


Figure 7-figure supplement 1.

Analyses of CD133⁺ vesicles

(A) Immunofluorescence on SKO liver 2 days after PHx. Asterisks indicates vasculature.

(B) Immunofluorescence on MCF10A cells treated with inhibitors.

(C) qRT-PCR analysis of PLC cell lysates after treatment with CD133⁺ vesicles isolated from MEK inhibitor-treated PLC cells. RNase and Triton X-100 were used to digest the RNA content of the vesicles. *P < 0.05, **P < 0.01 (two-tailed unpaired t test). n.s., not statistically significant. Means ± SD from 3 replicates are shown.

(D) EM image of immunogold staining on the isolated vesicles. CD133 (arrows) and HuR (arrowheads) were double stained using secondary antibodies conjugated with different sizes of gold particles.

Scale bars, 50µm (A and B) and 50nm (D).

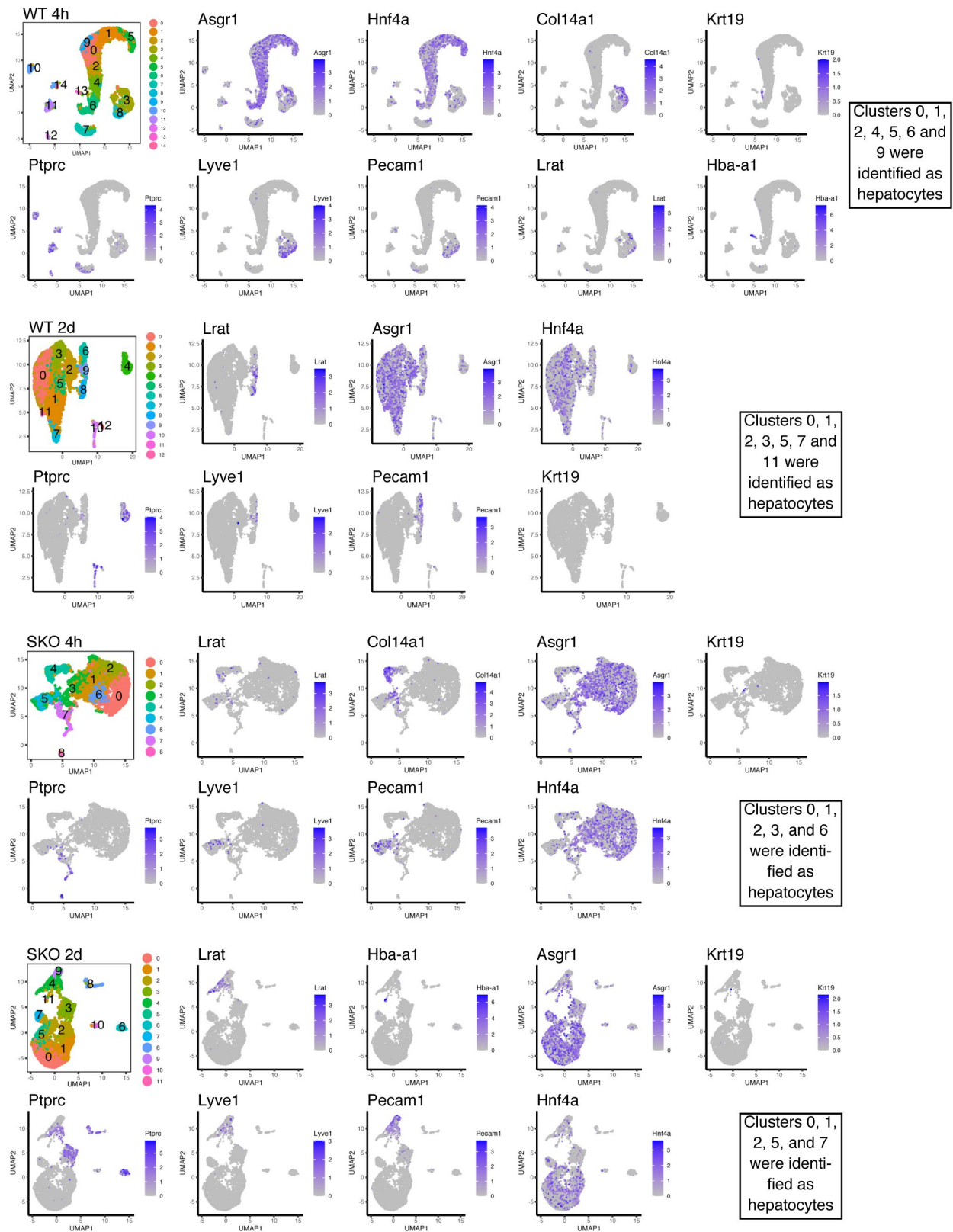


Figure 7-figure supplement 2.

Identification of hepatocytes in the scRNA-seq data set

UMAP analysis of the isolated cells from indicated livers. Hepatocyte markers and NPC markers were analyzed after the unsupervised clustering.

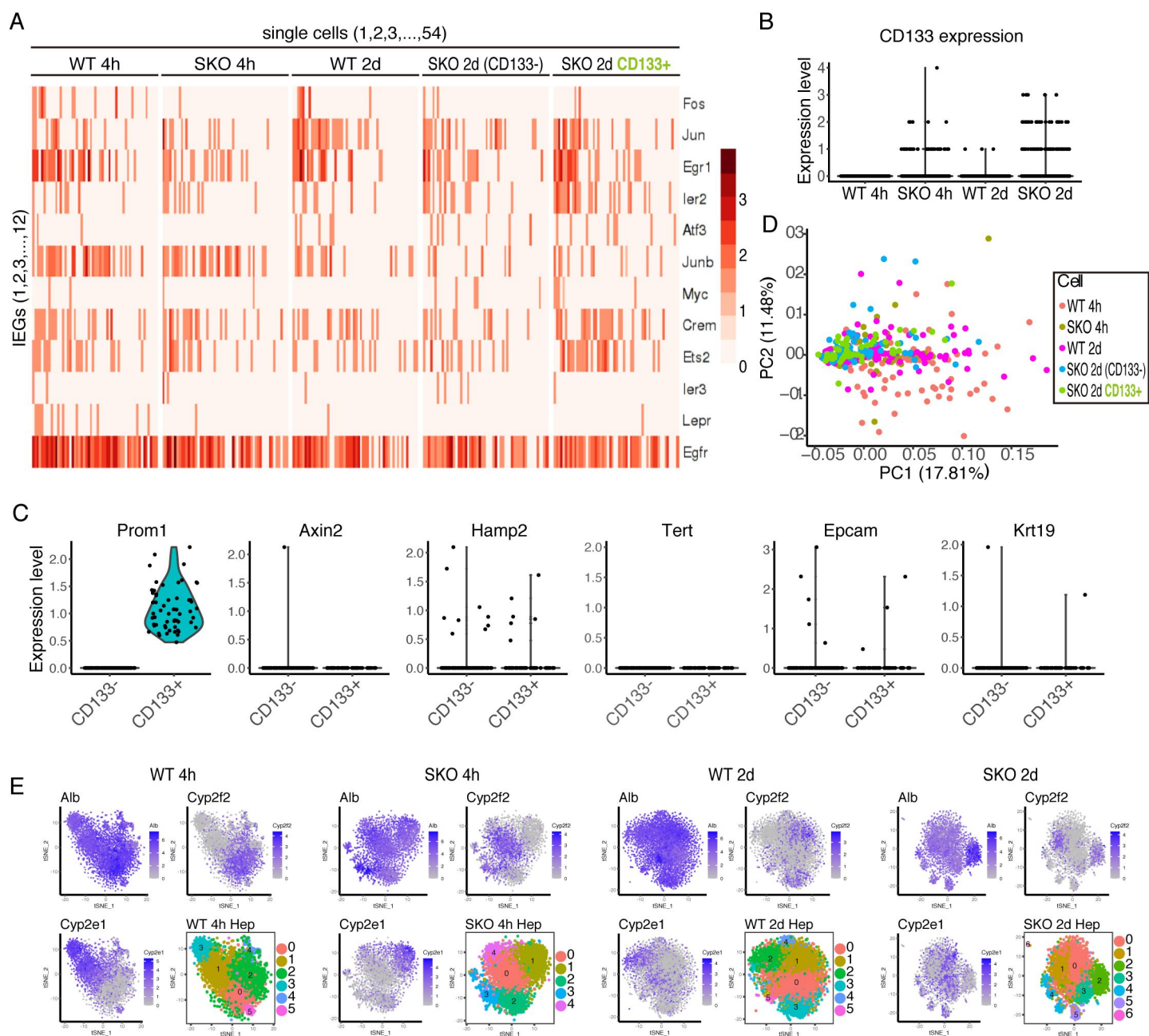


Figure 7-figure supplement 3.

Single cell RNA-seq analysis of WT and SKO livers after PHx

- (A) Heatmap analysis of single hepatocytes.
- (B) Detection of CD133 expression in the single cell RNA-seq data.
- (C) Expression levels of indicated hepatocyte subtype markers and stem cell-like markers. CD133- and CD133+ hepatocytes in the SKO liver 2 days after PHx were compared.
- (D) Principal component analysis with the IEGs.
- (E) tSNE analysis. Clusters likely reflect spatial locations within the tissue.

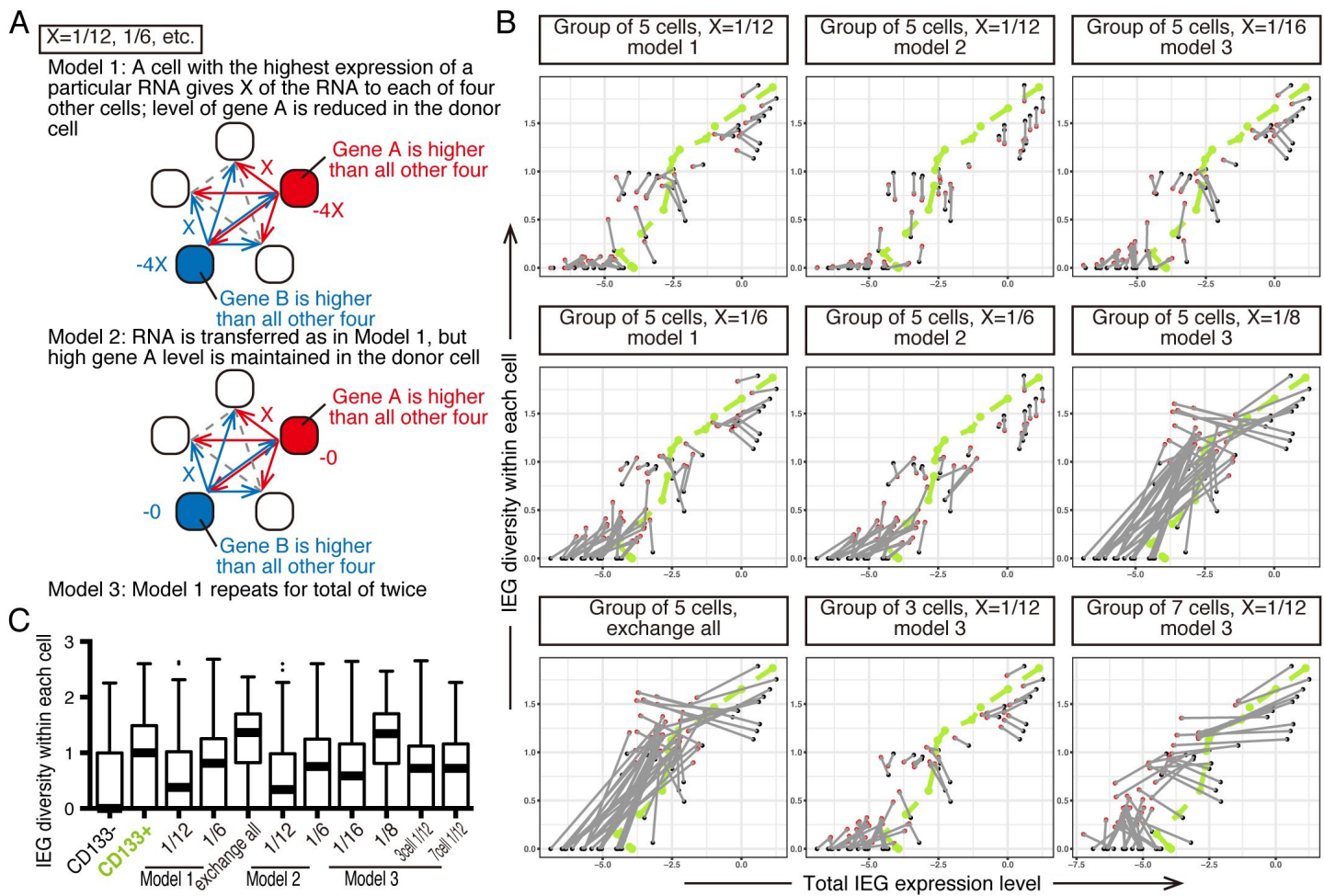


Figure 7-figure supplement 4.

Simulation analysis of the IEG exchange with scRNA-seq of the regenerating SKO liver

(A) Possible models tested for the simulation. Examples with 5 communicating cells are shown. X is a variable that reflects the amount of RNA exchanged between each pair of cells. (B and C) Simulation with different RNA exchange model, different variable X and cell numbers.

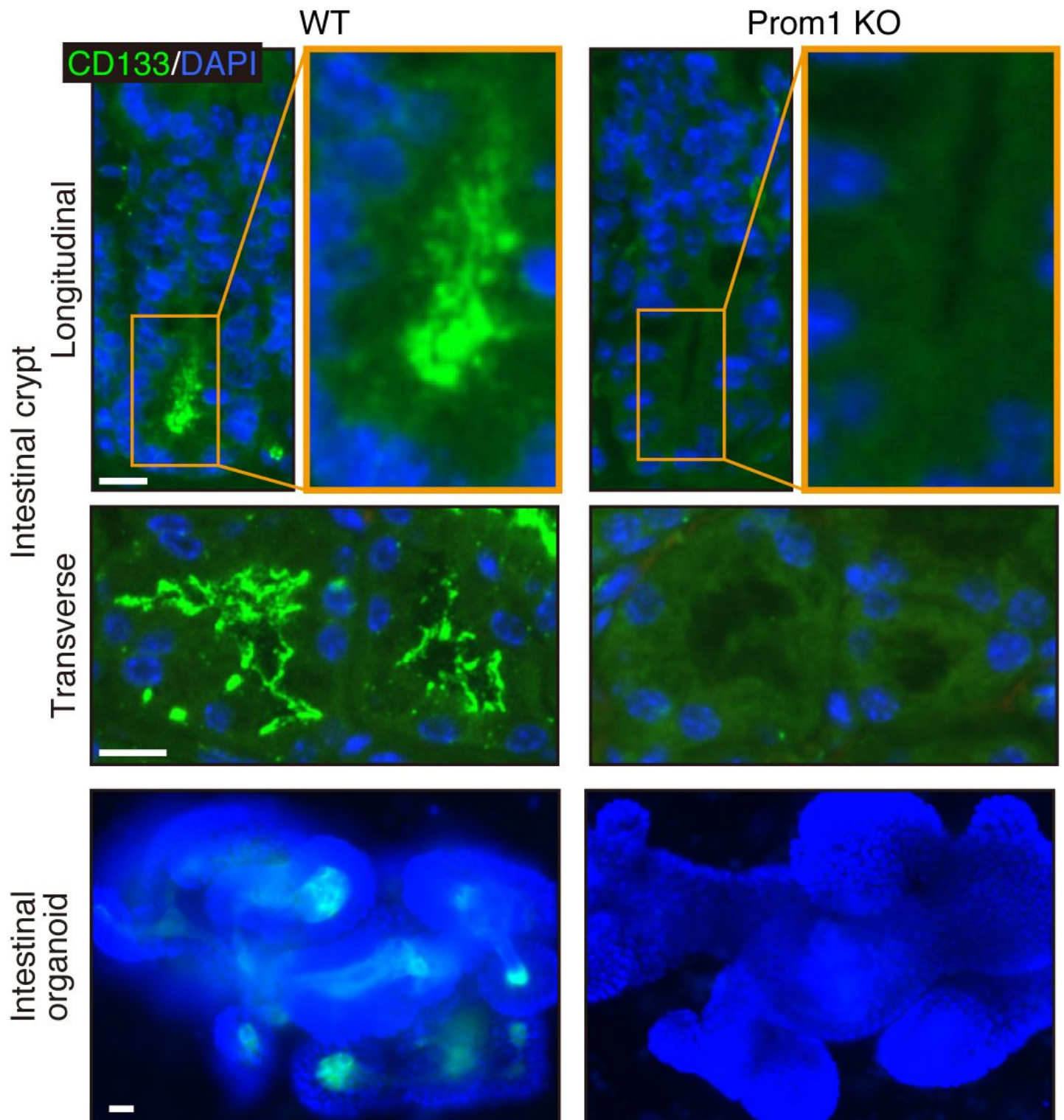


Figure 8-figure supplement 1.

CD133 is required for proliferation of crypt cells in the intestinal organoids during signal deficit

Immunofluorescence on WT and Prom1 KO mouse intestinal tissue sections and intestinal organoids. Scale bars, 20 μ m.

AAV Cre (AAV8.TBG.PI.Cre.rBG) was purchased from the Penn Vector Core and injected into the tail vein ($2 \times 10^{11}/28$ gbw). For chemical induction of hepatocellular carcinoma (HCC), DEN (25 mg/kg; N0258-1G; Sigma) was injected intraperitoneally into mice at postnatal day 15. BrdU (1 mg/mice; Sigma) was intraperitoneally injected twice a day for 3 days.

Cell culture and experiments

PLC/PRF/5 cells were cultured in DMEM medium supplemented with 10% fetal bovine serum. The MCF10A cells were cultured in DMEM-F12 medium (Gibco) with 5% horse serum (Gibco) and 1% penicillin-streptomycin supplemented with 20 ng/mL human EGF (Gemini), 0.5 µg/mL hydrocortisone, 100 ng/mL cholera toxin (Sigma-Aldrich), and 10 µg/mL human insulin (Gemini). The cultures were maintained at 37 °C in a humidified environment containing 5% CO₂. For inhibitors treatment, PLC/PRF/5 cells were treated with HGF and SHP099 (10 µM; CHEMIETEK; CT-SHP099) or trametinib (10 nM; APEXBio; A3018) for 2-6 days after cell plating and starvation overnight at the second day. MCF10A cells were treated with SHP099 (20 µM) or trametinib (10 nM) for 2-6 days after cell plating. Medium was changed every two days. Cells were transfected with CD133 Myc-Tag expression vector (HG15024-CM; SinoBiological) using Lipofectamine 3000 transfection reagent (Invitrogen). For lentivector transfection, the fusion protein and fluorescent proteins were subcloned into pLJM1 vector. HEK293T cells were transfected with the lentivectors together with psPAX2 and pCMV-VSV-G, using Lipofectamine 3000 (Thermo Fisher Scientific). Supernatant was collected, centrifuged at 3000g for 5min, filtered through 0.45µm filter and used to transfect the PLC cells. The lentiCRISPR v2 was used for the CRISPR experiment.

Immunostaining

Freshly frozen sections or fixed frozen sections were used for immunofluorescent staining. Freshly frozen sections and cultured cells were fixed by cold 4% PFA in PBS, cold acetone or cold methanol for 30 min to overnight, depending on the antibodies used. Antigen retrieval was performed for some stainings for 10 min to 1 hour, depending on the antibody used. Antibodies for Ki67 (14-5698-80; eBioscience), HNF4α (sc 8987; Santa Cruz Biotechnology), mouse CD133 (14-1331-82; eBioscience), human CD133 (86781; CST), E-cadherin (sc-7870; Santa Cruz Biotechnology), GFP (04404-84; Nacalai Tesque), Shp2 (sc-280; Santa Cruz Biotechnology), Porcupine (ab105543; abcam), β-catenin (sc-7199; Santa Cruz Biotechnology), CHMP2B (ab157208; abcam), HuR (ab200342; abcam, and 66549-1-Ig; proteintech), VE-Cad (AF1002; R&D SYSTEMS), BrdU (555627; BD Biosciences), PECAM (13-0311-81; eBioscience), HGF (ab83760; abcam), GFAP (Z0334; Dako), EpCAM (552370; BD Biosciences), Sox9 (AB5535; Millipore), CD44v6 (GTX75661; GeneTex) and AFP (AF5369; R&D SYSTEMS) were used as primary antibodies. Fluorescent images were taken with a monochrome camera, and pseudo colors were applied for each channel. STORM images were taken with Nikon A1R TIRF STORM Microscope at the Moores Cancer Center. Cryo-ultramicrotome sectioning, fluorescence staining, immunogold staining and electron-microscopy (EM) were performed by the Electron Microscopy Core Facility at UCSD.

In situ hybridization

Freshly frozen sections of the regenerating SKO liver and PLC cell line treated with Meki were subjected to HCR RNA-FISH, using probes and reagents purchased from Molecular Instruments after the CD133 staining. Freshly frozen sections were fixed with PFA for 30min, washed with PBS and further fixed/permeabilized with EtOH overnight. PLC was fixed with MeOH overnight. The fixed sections and cells were then immunostained for CD133 using the antibodies described above. After the immunostaining, HCR RNA-FISH was performed according to the protocol provided by Molecular Instruments. ProtectRNA RNase Inhibitor (R7397; SIGMA) was added to all the reagents used. The images were acquired using Leica SP8 confocal microscopy.

Primary hepatocyte isolation and culture

Each liver was perfused with 10 ml of HBSS (without Ca^{2+} or Mg^{+}) containing 0.5 mM EDTA, 4 ml of HBSS (without Ca^{2+} or Mg^{+}) and 10 ml of collagenase solution [2 mg/ml collagenase H (11074059001; Roche), 0.1 mg/ml DNaseI (10104159001; Roche), 5 mM CaCl_2 and 0.9 mM MgCl in HBSS] sequentially. All the solutions were pre-warmed to 37 °C before use. Livers were dissected and further incubated in 7 ml of collagenase solution at 37 °C for 5 min and dissociated. Dissociated cells were passed through cell strainer and centrifuged at 50 g for 5 min to precipitate hepatocytes. For qRT-PCR, the pellet was immediately dissolved in Trizol for the hepatocyte fraction. The supernatant containing NPCs was washed with PBS at 50 g for 5 min three times to eliminate hepatocytes, and precipitated at 400 g for 5 min and dissolved in Trizol and used as NPC fraction. For hepatocyte culture, the hepatocyte pellet after the first centrifugation was resuspended and washed with PBS at 50 g for 5 min three times to eliminate NPCs, resuspended in the culture media [Williams' Media E supplemented with Primary Hepatocyte Maintenance Supplements (CM4000; Gibco) as instructed] and seeded on collagen-coated plates.

Intestinal organoid culture

After dissecting the intestinal tube from the mice, the tubes were washed with PBS, by flushing with syringe. The villi were gently scraped off and the tissues were cut to smaller pieces. The tissue fragments were rinsed with PBS three times, and then incubated in PBS with 1mM EDTA for 30 minutes. After the incubation, the fragments were put in fresh PBS and rigorously shaken, and the cells/debris that came off were discarded. The tissue fragments were then incubated in PBS with 5mM EDTA for one hour. After the incubation, the fragments in the solution were rigorously shaken and pipetted, and the cell clumps that came off were used as the isolated crypts. The crypts were passed through 100 μm cell strainer and spun down and washed with PBS at 300 x g for 10 minutes three times. The isolated crypt fragments were embedded in Matrigel and cultured in DMEM/F12 containing B27 (17504044; Gibco), N2 supplement (17502048; Gibco), 1.25 mM n-Acetylcysteine, 50 ng/ml EGF, 100 ng/ml Noggin and 500 ng/ml R-spondin 1. Organoids were passaged in the same media, and treated with 50 nM trametinib.

Immunoblotting

Proteins were extracted in RIPA buffer, and immunoblotting was performed using standard protocols. In addition to antibodies used for immunostaining, antibodies for EpCAM (bs-1513R; Bioss), HNF4 α (GTX89532; GeneTex), GAPDH (5174; CST), β -actin (A5316; Sigma), Stat3 (9132; CST), Erk1/2 (4695; CST) and Cyclin D1 (sc 20044; Santa Cruz Biotechnology) were used as primary antibodies.

Hydrodynamic tail vein injection

Plasmids pT3-elongation factor 1 α (EF1 α)-Shp2 (5 $\mu\text{g}/\text{ml}$), pT3-EF1 α -GFP (5 $\mu\text{g}/\text{ml}$), p-cytomegalovirus (pCMV)-sleeping beauty (SB) transposase (1.5 $\mu\text{g}/\text{ml}$), or were diluted in PBS. Total volume of 2 ml/20 g.b.w. was rapidly injected into tail vein. The GFP-expressing vector has transposon sequences, and when injected together with the SB transposase-expressing vector, the GFP expressing sequence was integrated into the genome. Plasmids were prepared using the GenElute HP Endotoxin-Free Plasmid Maxiprep Kit (NA0410; Sigma). Silica fines in the final product were carefully eliminated by additional centrifugation at 12,000 g for 30 min.

Isolation of CD133⁺ vesicles

Liver perfusion was similar to hepatocyte isolation as described above, except that collagenase Type IV (17104019; Gibco) was used instead of collagenase H to minimize unspecific digestion of proteins other than collagen. The livers were dissected, immersed in the collagenase solution and dissociated and mildly lysed. Cells and debris were removed by centrifugation at 4,000 g for 10 min three times, followed by filtration with 0.45 μm PES filter. We obtained around 2 ml of

solution per mouse at this point. EDTA was added to suppress the collagenase activity and sodium azide was added to prevent possible degradation by bacterial growth. For PLC cells, cells were lysed in PBS by Dounce homogenizer, and cells and debris was removed similarly. FITC conjugated anti-CD133 antibody (1:500 dilution; 11-1331-82; Invitrogen) was added, and the solution was incubated at 4 °C overnight. Objects labeled with anti-CD133 antibody were then targeted with magnetic particles, using FITC selection cocktail (18558; STEMCELL TECHNOLOGY), and isolated using MACS MS column (130-042-201; Miltenyi Biotec). CD133⁺ fraction was directly eluted with either lysis buffer for immunoblotting or lysis buffer for RNA isolation using RNeasy Plus Micro Kit (74034; QIAGEN). Isolation protocol for total RNAs including small RNAs was used according to the manufacturer's instruction with the RNeasy Plus Micro Kit. For the agarose gel electrophoresis, total RNAs were normalized to same concentrations, mixed with RNA loading dye (final concentration of 47.5% formamide, 0.01% SDS, 0.01% bromophenol blue, 0.005% Xylene Cyanol FF, 0.01% SYBR safe, 0.5 mM EDTA), denatured at 65 °C for 15 min, cooled on ice, and loaded to the gel. For treatment of cells with the isolated vesicles, Anti-PE MultiSort Kit (130-090-757; Miltenyi) was used instead of FITC selection cocktail, to enzymatically detach the vesicles from the magnetic beads. As the vesicles were not likely secreted and endocytosed naturally, we forced the cells to take the vesicles by using Lipofectamine 3000.

Isolation of extracellular vesicles

Extracellular vesicles were isolated from the culture supernatant using Total Exosome Isolation Reagent (Invitrogen). Debris or cell contaminants were removed by centrifugation, larger vesicles such as apoptotic bodies were subsequently removed by filtration, and extracellular vesicles were precipitated according to the manufacturer's instruction. Total RNAs were extracted in the same way as the extraction from CD133⁺ vesicles described above.

qRT-PCR

RNAs extracted by either Trizol or RNeasy Plus Micro Kit were reverse transcribed to cDNAs using High-Capacity cDNA Reverse Transcription Kit (4368814; Applied Biosystems), and quantitative PCR was performed using DyNAmo Flash SYBR Green qPCR Kit (F415S; Thermo Scientific). Primer sequences are listed in supplementary Table S1.

Single cell RNA library construction and sequencing

Single hepatocytes were isolated from WT at 4 hrs (4h) and 2 days (2d), SKO at 4h and 2d, and DKO at 2d after PHx, by a two-step in-situ perfusion with collagenase H as described above. To remove dead cells and debris, the pelleted hepatocytes were resuspended in 45% Percoll (sigma-aldrich) and centrifuged at 50g for 10 min. Cells were further washed with PBS and counted with hemocytometer. The isolated hepatocytes were loaded onto a 10x Chromium Controller and then partitioned into nanoliter-scale Gel Beads-In-Emulsion (GEMs). The volume of single cell suspension was calculated in order to generate 5000 GEMs per sample. Libraries were constructed with Chromium Single Cell 3' Reagent Kits (v3 chemistry, 10x Genomics). Sequencing was performed using HiSeq 4000 (Illumina) at IGM Genomics Center, University of California, San Diego, with the following read length: Read 1, 28 bp, including 16 bp cell barcode and 12 bp unique molecular identifier (UMI); Read 2, 98 bp transcript insert; i7 sample index, 8 bp. For the single cell RNA-seq data, GEO accession number: GSE169320

Single cell RNA-seq data preprocessing

The paired reads from WT and SKO samples were aligned to mouse reference genome GRCm38 using CellRanger package (v3.0.2). And the reads from DKO were aligned to modified GRCm38, with Prom1-CreERT2 sequence manually added. An expression matrix, with each row representing a gene and each column representing a cell, was generated for each sample, and then filtered for downstream analysis. In brief, for each sample, genes expressed in less than 3 cells were removed; cells failed to meet following criteria were removed: 1) the number of genes

detected in each cell should be over 200 and less than 5000; 2) the UMIs of mitochondrial genes should be less than 20% of total UMI; 3) the UMIs of a cell should be less than 20,000 to avoid non-singlet (i.e. transcriptome representing more than one cell).

Detection of cell subpopulations

To identify subpopulations within hepatocytes and select clusters for further analyses, we examined each sample using R package Seurat (v2.4) (Butler et al, 2018). First, the raw expression matrix was normalized by the total expression, multiplied by scale factor 10,000, and log-transformed. Next, we regressed on the number of UMIs per cell as well as mitochondrial gene percentage. After normalization and scaling, z-scored residuals were stored for dimensionality reduction and clustering (Figures S6). The hepatocytes showed high expression of *Hnf4a* and *Asgr1*. And other clusters with expression of *Ptprc*, *Lyve1*, *Pecam1*, *Lrat*, *Hba-a1*, and *Krt19* (contaminant immune cells, endothelial cells, hepatic stellate cells, and erythrocytes) were removed. Next, for hepatocytes in each sample, PCA was performed first, and 75 PCs were used for tSNE analysis with other parameters, including perplexity = 30 and maximum iteration = 2000. To find hepatocyte subpopulations, the same PCs were imported into FindClusters from Seurat. All samples were also colored by *Alb* (marker for hepatocytes), *Cyp2f2* (marker for periportal hepatocytes) and *Cyp2e1* (marker for pericentral hepatocytes) expression. CD133⁺ cells of SKO liver (2d after PHx) were not limited to, but abundantly identified in one of the two clusters of periportal hepatocytes (80%), which was consistent with our observations on tissues of the PHx model. After comparing differentially expressed genes in two periportal clusters, we found the cluster enriched with CD133⁺ cells in SKO 2d sample, had higher *Alb* and *Hamp* expression, presumably reflecting a population that is located in a particular zone within the periportal area. The same population (*Alb*^{high}, *Hamp*^{high}, *Cyp2f2*⁺) was also observed in other samples. For fair comparison, further analysis was limited to this periportal population to avoid variances due to spatial locations within the liver tissue.

Calculation of total IEGs' expression levels

To obtain the IEG list for analysis, we checked data in public database (<http://software.broadinstitute.org/gsea/msigdb/index.jsp>) and publications (Locker et al, 2003; Su et al, 2002), for genes expressed during liver regeneration or after liver injury. Among the genes upregulated, we eliminated the genes that were irrelevant to cell growth or those acting as negative feedback regulator. As a result, we obtained 12 IEGs for further analyses, including *Fos*, *Jun*, *Egr1*, *Ier2*, *Atf3*, *Junb*, *Myc*, *Crem*, *Ets2*, *Ier3*, *Lepr*, *Egfr*. To integrate the total expression level of the 12 genes, we defined a score based on zero-inflated Negative Binomial (ZINB) model. ZINB regression is for modeling count variable with excessive zeros. It was first introduced to model single-cell RNA-seq data by Miao et al. (Miao et al, 2018), and it had shown high accuracy in differential expression analysis compared to other models (Wang et al, 2019). The score for total IEGs expression was computed as follows. 1) Within a group of cells, we first fitted counts of each gene by a ZINB model:

$$f_{ZINB}(n; \theta, r, p) \equiv P(N = n) = \theta \cdot I(n = 0) + (1 - \theta) \cdot f_{NB}(n; r, p)$$

$$= \theta \cdot I(n = 0) + (1 - \theta) \cdot \binom{n + r - 1}{n} p^n (1 - p)^r$$

where θ is a proportion of 'real' zeros in this group of cells ('real' zeros represent that the gene was not expressed in some cells, instead of the dropout caused by mRNA capture procedure); r and p were parameters from negative binomial distribution, representing size and probability, respectively; n is the median normalized count (Miao et al., 2018) of genes. ZINB model connects

‘real’ zeros and a negative binomial model with θ and indicator function I . The observed excessive zeros from single-cell RNA-seq dataset might come from either ‘real’ zeros or zeros in negative binomial model. 2) Therefore, the expected count or mean for each gene was calculated as follow:

$$\begin{aligned}\mu_{ZINB} &= P(\text{gene not expressed}) \cdot 0 + P(\text{gene expressed}) \cdot \mu_{NB} \\ &= (1 - \theta) \cdot \frac{pr}{1 - p}\end{aligned}$$

For each group of cell c and each gene g , we calculated the expected count as $\mu_{c,g}$. With the same method, the expected count of gene g among all cells used for this analysis was calculated as $\mu_{a,g}$. 3) For cell cluster c , the score of total IEGs expression level was defined as:

$$S_c = 2^{\frac{1}{N_g} \sum_g \log_2 \left(\frac{\mu_{c,g} + 0.001}{\mu_{a,g}} \right)}, \text{ where } N_g = 12$$

4) For a single cell in a group, a , for example, we calculated its binary distance to all other cells and selected 20 closest cells as a subgroup, called ca . The score for ca was calculated following step 1) to 3) as described above and plotted as a dot in **Figure 7H** [↗](#).

IEG diversity within each cell and variation among cells

We first used Shannon’s entropy of all cells in a group to indicate the IEG diversity. The entropy E_j of cell j was computed as:

$$\begin{aligned}E_j &= -\sum_{i=1}^N p_{i,j} \log_2 p_{i,j} \\ p_{i,j} &\equiv r_{i,j} / \sum_i r_{i,j} \\ r_{i,j} &\equiv \text{count}_{i,j} / \mu_i\end{aligned}$$

where $\text{count}_{i,j}$ represents the median normalized count of gene i in cell j , and μ_i equals the expected count among all cells used for analysis. The higher entropy defined above practically reflects 1) more species of IEGs are expressed by a cell, as well as 2) less overall deviation from mean expression values of IEGs within a cell. To describe the variation among cell subpopulation, we used cell pair-wise Euclidean distance with normalized and scaled data by Seurat. However, in order to do fair comparison between samples when using scaled data, we first integrated four datasets by Canonical Correlation Analysis (CCA) with Seurat, before calculating the distance. Finally, the significance of diversity and variation between cell groups were decided by Wilcoxon rank sum test and adjusted by FDR.

RNA exchange simulation

We plotted IEG diversity (entropy) against total IEG expression levels in SKO-2d sample. The cells were first divided into CD133⁻ and CD133⁺ groups. Within each group, all cells were ordered by their entropies from low to high.

Starting from the first cell, we applied a moving average to entropies, with a sliding window of 20 cells and a step size of 5. Within each window, we also computed the total IEG expression value as described above. We then plotted IEG entropy against total expression level for CD133⁻ and CD133⁺ groups. Every sliding window with 20 single cells was treated as a meta-cell. Next, we examined how the cells would behave in the plot, when simulating the RNA exchange among CD133 cells. The basic idea was to randomly divide all CD133⁻ cells into small groups with size N and make cells within the same group exchange IEGs with each other. We tested four possible RNA exchange models (**Figure 7-figure supplement 3A** [↗](#)):

- Model 1: Within a small group of size N , for each IEG, the cell with the highest expression gives X of its RNA to each of the other $(N-1)$ cells; the corresponding RNA level is reduced by $(N-1)*X$ in the donor cell.

$$\begin{cases} N = 3, 5, 7 \\ X = \frac{1}{16}, \frac{1}{12}, \frac{1}{8}, \frac{1}{6}, \frac{3}{16}, \frac{1}{4} \end{cases}$$

- Model 2: Within a small group of size N , for each IEG, RNA is transferred as described in Model 1, but RNA level is maintained in the donor cell because of its continuous transcription.

$$\begin{cases} N = 3, 5, 7 \\ X = \frac{1}{16}, \frac{1}{12}, \frac{1}{8}, \frac{1}{6}, \frac{3}{16}, \frac{1}{4} \end{cases}$$

- Model 3: Within a small group of size N , for each IEG, there are two rounds of RNA exchange. Considering that the CD133 expression was already detected (in RNA level) at 4 hours after PHx (**Figure 7-figure supplement 2B** [↗](#)), this is likely the reasonable model, as they should have multiple chances of mRNA exchange while they change their profile along the time until 2 days after PHx. The first round is the same as Model 1: recipients get X_1 of the RNA in a donor cell

and the donor cell loses $(N-1) \cdot X_1$. After the first round of exchange is finished, the cell with highest expression now is designated as a new donor. During second round, new donor loses $(N-1) \cdot X_2$ of its RNA and each of the other $(N-1)$ cells get X_2 .

$$\left\{ \begin{array}{l} N = 3, 5, 7 \\ X_1 = \frac{1}{16}, \frac{1}{12}, \frac{1}{8}, \frac{1}{6}, \frac{3}{16}, \frac{1}{4} \\ X_2 = \frac{1}{16}, \frac{1}{12}, \frac{1}{8}, \frac{1}{6}, \frac{3}{16}, \frac{1}{4} \end{array} \right.$$

- Model 4 (exchange all): Within a small group of size N ($N = 3, 5, 7$), for each IEG and each cell, the RNA levels become the same, which equal to the mean values before exchange.

After the RNA exchange simulation was applied, the IEG diversity (entropy) and total IEG expression were re-calculated in order to track their changes. For [Figure 7M](#), the IEG diversity was calculated as described above, and the analysis included both Cyclin D1 negative and -positive cells.

Bulk RNAseq sample preparation and data processing

Total RNAs extracted from vesicles as described above were subjected to ribosome-depleted total RNA library prep (Illumina) at UCSD Genomics Core. Sequencing was performed at UCSD Genomics Core on the NovaSeq 6000 platform. Raw reads data quality control was performed using fastqc, sequenced reads were mapped to the hg19 reference genome using STAR, and the number of mapped reads to each gene was counted by htseq-count. The R package DESeq2 was then used to normalize raw reads and perform differential expression analysis between CD133⁺ vesicles vs whole cell samples, as well as EVs vs whole cell samples. The significantly expressed genes were determined with adjusted p value < 0.05.

Heatmap

For single cell RNA-seq, we randomly selected 54 cells from the same periportal cluster as described above, to make the heatmap. The raw counts were log transformed with pseudo-count = 1. For bulk RNA-seq, we made two heatmaps to compare gene expression between CD133⁺ vesicles and EVs. In [Figure 6D](#), the variance stabilizing transformed values of interested genes from CD133⁺ vesicles and EVs were extracted and scaled. In Figure supplement 5E, the transformed values from CD133⁺ vesicles and EVs were further standardized to whole cells.

Statistical analysis

All statistical analyses of data are described in the corresponding figure legends. All statistical analyses were performed with GraphPad Prism 7, except R was used for the single cell RNA-seq analyses.

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Reviewer #1 (Public Review):

This study was designed to examine the bypass of Ras/Erk signaling defects that enable limited regeneration in a mouse model of hepatic regeneration. The authors show that this hepatocyte proliferation is marked by expression of CD133 by groups of cells. The CD133 appears to be located on intracellular vesicles associated with microtubules. These vesicles are loaded with mRNA. The authors conclude that the CD133 vesicles mediate an intercellular signaling pathway that supports cell proliferation. These are new observations that have broad significance to the fields of regeneration and cancer.

The primary observation is that the limited regeneration observed in livers with Ras/Erk signaling defects is associated with CD133 expression by groups of cells. The functional significance of CD133 was tested using Prom1 KO mice - the data presented are convincing.

The major weakness of the study is that some molecular mechanistic details are unclear - this is, in part, due to the extensive new biology that is described. Nevertheless, the data used to support some key points in this study are unclear:

a) What is the evidence that the observed CD133 groups of cells are not due to clonal growth. Is this conclusion based on the time course (the groups appear more rapidly than proliferation) or is this based on the GFP clonal analysis?

b) What is the evidence that the CD133 vesicles mediate intercellular communication. This is an exciting hypothesis, but what is the evidence that this happens? Is this inferred from IEG mRNA diversity? or some other data. Is there direct evidence of transfer - for example, the does the GFP clonal analysis show transfer of GFP that is not mediated by clonal proliferation? Moreover, since the hepatocytes are isogenic, what distinguishes the donor and recipient cells?

Increased clarity concerning what is hypothesis and what is directly supported by data - would improve the presentation of this study.

Reviewer #2 (Public Review):

The manuscript by Kaneko set out to understand the mechanisms underlying cell proliferation in hepatocytes lacking Shp2 signals. To do this, the authors focused on CD133 as the proliferating clusters of cells in the Shp2 knockout (SKO) livers are CD133 expressing. After excluding the contribution of progenitors that are CD133 to this cell population, the authors focused on the intrinsic regulation of CD133 by Met/Shp2 regulated Ras/Erk pathway and showed upregulation of CD133 to be a compensatory signal to overcome loss of Ras/Erk signal and suggested Wnt10a in the regulation of CD133 signal. The study then focused on the observed filament localization of CD133 in the CD133+ cluster of cells. The study went on to identify the CD133+ vesicles that contain primarily mRNA vs. microRNA like other EVs. Specifically, the authors identified several mRNA species that encode IEGs, indicating a potential role for these CD133+ vesicles in cell proliferation signal transmission to neighboring cells via delivery of the IEG mRNAs as cargos. Finally, they showed that the induction of CD133 (and by derivative, the CD133+ vesicles) are necessary for maintaining cell proliferation in the cell cluster with high proliferation capacities in the SKO livers; and in intestinal crypt organoids treated with Met inhibitors to block Ras/ERK signal. In the revised manuscript, the authors more definitively identified the CD133+ vesicles. The authors also provided additional experimental evidence demonstrating the role of these CD133+ vesicles in cell-cell communication. The functional significance of CD133 on this cell-cell communication was further demonstrated with genetic knockout studies.

Author Response

The following is the authors' response to the original reviews.

eLife assessment

This important study was designed to examine the bypass of Ras/Erk signaling defects that enable limited regeneration in a mouse model of hepatic regeneration. This hepatocyte proliferation is associated with the expression by groups of cells of mRNA-loaded CD133+ intracellular vesicles that mediate an intercellular signaling pathway that supports proliferation. These are new observations, supported by convincing data, that have broad significance to the fields of regeneration and cancer.

First of all, we greatly appreciate the very positive take of this work by eLife editors and also thank the two reviewers for their constructive comments. We have provided point-by-point responses as follows.

Reviewer #1 (Public Review):

This study was designed to examine the bypass of Ras/Erk signaling defects that enable limited regeneration in a mouse model of hepatic regeneration. The authors show that this hepatocyte proliferation is marked by expression of CD133 by groups of cells. The CD133 appears to be located on intracellular vesicles associated with microtubules. These vesicles are loaded with mRNA. The authors conclude that the CD133 vesicles mediate an intercellular signaling pathway that supports cell proliferation. These are new observations that have broad significance to the fields of regeneration and cancer.

The primary observation is that the limited regeneration observed in livers with Ras/Erk signaling defects is associated with CD133 expression by groups of cells. The functional

significance of CD133 was tested using Prom1 KO mice - the data presented are convincing.

The major weakness of the study is that some molecular mechanistic details are unclear - this is, in part, due to the extensive new biology that is described. Nevertheless, the data used to support some key points in this study are unclear:

We fully agree that some details of the molecular mechanisms are yet to be elucidated for the CD133⁺ vesicles (intercellsomes, as we named). This is the first report of a new direct cell-cell communication mechanism provoked in stress response to proliferative signal deficit.

Remarkably, many questions remain open for the molecular mechanisms for formation and functions of relatively well-characterized structures such as exosomes/EVs, despite a huge body of literature since their discoveries.

a) What is the evidence that the observed CD133 groups of cells are not due to clonal growth. Is this conclusion based on the time course (the groups appear more rapidly than proliferation) or is this based on the GFP clonal analysis?

This is indeed a very critical point for this study. Our initial thought and efforts were indeed on finding evidence that supports clonal expansion of progenitor cells. However, the experiments showed that the CD133⁺ cells were negative for all other stem/progenitor cell markers and that they are mature hepatocytes. CD133 expression was upregulated dramatically in regenerating livers and disappeared upon completion of liver regeneration. Furthermore, suppression of Ras-Erk signaling by Shp2 and Mek inhibitors robustly induced CD133 expression in a variety of cancer cell lines in culture in vitro.

At 2 days after PHx, we already observed big colonies, which were unlikely derived from a single initiating cell (Figure 1). The GFP clonal analysis unambiguously demonstrated the heterogenous origin of the clustered cells (Figure 3). We detected mixed GFP-positive and -negative cells within each colony, without a single colony consisting entirely of GFP-positive cells. The original colony sizes were estimated to be 10 cells or more (Figures 3G and Figure 3-figure supplement 1B). Thus, both the sizes and compositions in the GFP clonal analyses support the assertion that CD133⁺ cell clusters originated from multiple mature hepatocytes.

b) What is the evidence that the CD133 vesicles mediate intercellular communication. This is an exciting hypothesis, but what is the evidence that this happens? Is this inferred from IEG mRNA diversity? or some other data. Is there direct evidence of transfer - for example, the does the GFP clonal analysis show transfer of GFP that is not mediated by clonal proliferation? Moreover, since the hepatocytes are isogenic, what distinguishes the donor and recipient cells? Increased clarity concerning what is hypothesis and what is directly supported by data - would improve the presentation of this study.

Per the reviewer's advice, we have clarified these points in the revised version. Our proposal that CD133 vesicles mediate intercellular communication was supported by these experimental results.

A). Data in Fig. 5 suggest direct trafficking of the vesicles, as CD133 existed on the filaments that bridge the tightly contacting cells. This was confirmed by two different CD133 antibodies in mouse and human. Of note, CD133⁺ vesicles are negative for CD9, CD63 or CD81, markers for exosomes/EVs. We could only isolate CD133⁺ vesicles from cell lysates in vitro and mouse tissue lysates, but not from cell supernatants from which exosomes/EVs are isolated.

B). More direct evidence of the transfer was presented in Fig. 6H, showing Myc-tagged CD133 molecules transferred from one cell to another. In response to reviewers' comments, we now

conducted correlative light and electron microscopy to characterize the exchange event around the cell-cell border at EM level (new Figure6-figure supplement 2).

C). Further experimental evidence was provided in the single and double gene KO experiments in Fig. 8E-G, suggesting the functional significance of CD133 in intercellular communication.

D). In addition to the data above, the IEG mRNA diversity analyses based on scRNA-seq support the mRNA exchange model. The isogenic CD133+ SKO hepatocytes were found to lack different IEG transcripts randomly. This is why we propose a mutually sharing model, rather than a donor and recipient model. Importantly, the mRNA diversity (entropy) model also illustrates the association of CD133 and “stemness”, as described in the discussion.

In sum, we believe that a most reasonable interpretation of the current data set is a model of direct cell-cell communication via CD133+ vesicles. We take the reviewer’s point and have made changes to the text to better distinguish conclusion and hypothesis, which will be validated in future studies.

Reviewer #2 (Public Review):

The manuscript by Kaneko set out to understand the mechanisms underlying cell proliferation in hepatocytes lacking Shp2 signals. To do this, the authors focused on CD133 as the proliferating clusters of cells in the Shp2 knockout (SKO) livers are CD133 expressing. After excluding the contribution of progenitors that are CD133 to this cell population, the authors focused on the intrinsic regulation of CD133 by Met/Shp2 regulated Ras/Erk pathway and showed upregulation of CD133 to be a compensatory signal to overcome loss of Ras/Erk signal and suggested Wnt10a in the regulation of CD133 signal. The study then focused on the observed filament localization of CD133 in the CD133+ cluster of cells. The study went on to identify the CD133+ vesicles that contain primarily mRNA vs. microRNA like other EVs. Specifically, the authors identified several mRNA species that encode IEGs, indicating a potential role for these CD133+ vesicles in cell proliferation signal transmission to neighboring cells via delivery of the IEG mRNAs as cargos. Finally, they showed that the induction of CD133 (and by derivative, the CD133+ vesicles) are necessary for maintaining cell proliferation in the cell cluster with high proliferation capacities in the SKO livers; and in intestinal crypt organoids treated with Met inhibitors to block Ras/Erk signal.

- 1. The identification of CD133+ vesicles is largely based on staining and costainings. Though the experiments are very well done with many controls and approaches, the authors may want to perform one or two key experiments with EM to definitively demonstrate the colocalization. For example, the mCherry experiment in Fig6H and the colocalization experiments for CD133 and HuR in Fig 7.*

Many thanks for the suggestion. We now completed the two suggested key experiments with new results added to the revised manuscript. For the mCherry experiment, we conducted correlative light and electron microscopy to characterize the exchange event between cells that stably express CD133-GFP fusion protein and mCherry+ cells (new Figure 6-figure supplement 2). The CD133-GFP was clearly found in the mCherry+ cells around the border, demonstrating the intercellular traffic. For the colocalization of CD133 and HuR, we performed double immunogold staining on the isolated vesicles, with the new results presented in the revised Figure7-figure supplement 1D.

- 1. Since CD133+ marks the 50nM intracellsome defined by the authors, it is unclear what the CD133- vesicles used as controls are. Are they regular EVs that are larger in size? This*

needs better clarification as they are used as a control for many experiments such as Fig 7A.

Per the advice, we added more explanation to the revised text. We used regular EVs as the control, since they are the well-studied intercellular communication vesicles. Since the EVs are highly heterogenous, we did not choose to select a specific subpopulation of EVs. We used the well-established polymer-based precipitation method to isolate the EV fraction from cell culture supernatant for RNA-seq analysis. We did detect the enrichment of micro-RNAs in the isolated EVs, consistent with reports in the literature. Strikingly, the CD133 vesicles isolated from cell lysates showed a completely distinct RNA profile, relative to the EVs.